

KALASIN STARCH PLANT - JET COOKER PROJECT

STANDARD INSTRUMENT HOOK-UP & TYPICAL ELECTRICAL LOOP DRAWINGS

Based on Equipment Packages in 11-instrument Manual/ & Master I/O Mapping Rev 3.6

DRAWING PACKAGE OBJECTIVES & ENGINEERING STANDARDS

Project Scope:	Jet Cooker Slurry & Flash Tank Instrumentation
Client / Owner:	INGREDION (THAILAND) CO., LTD.
Plant Location:	Kalasin Starch Manufacturing Plant, Thailand
System Integrator:	AEC INDUSTRIAL ENGINEERING CO., LTD.
Covered Packages:	23 Complete Equipment Types (CS-DS-01 to CS-DS-22)
Drawing Contents:	Piping Hook-Up, Loop Wiring Schematic & BOM
Electrical Standard:	Rockwell ControlLogix 1756 Architecture (24V DC / 4-20mA)
Hazardous Areas:	Ex ia / Ex d Classifications (937T Barriers & Glands)
CAD Standard:	1:1 Metric Millimeters (AutoCAD DXF R2010 Layered)
Piping Standard:	ASME B16.5 Cl.150/300 RF Flanges & NPT Threads
Document Status:	APPROVED ENGINEERING TEMPLATES (Rev 3.6)

DRAWING SYMBOLOGY & CAD LAYER CONVENTIONS

Process Piping:
Heavy solid line (0.7mm), showing process flow direction
Impulse Tubing:
Medium dashed line, 1/2" OD seamless SS316 tubing
Instrument Air:
Pneumatic air supply line, 6.0 bar dry filtered air
Analog Signal (AI/AO):
4-20mA HART loop, shielded twisted pair (Belden 8761)
Discrete Signal (DI/DO):
24V DC wetted contact or solenoid drive (Belden 9533)
IS Barrier (Ex ia):
Sky Blue block, Rockwell 937T Series Intrinsically Safe
Interposing Relay:
Orange block, Allen-Bradley 700-HL Series 24VDC Relay
Terminal Strip:
Green circle/box, Marshalling Panel CA1 (C#S##-X1)
Earth Grounding:
Protective Earth (PE) & Instrument Isolated Clean Earth (IE)

PREPARED BY

AEC INDUSTRIAL ENG.

Lead Instrumentation Eng.

CHECKED BY

AEC AUTOMATION LEAD

System Architect & PE

VERIFIED BY

PROJECT ELECTRICAL PE

Licensed Professional Eng.

APPROVED BY

INGREDION PROJECT MGR.

Client Automation Lead

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

INSTRUMENT HOOK-UP & LOOP DRAWING DIRECTORY
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DWG NO: KAL-JC-TYP-000
REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

PROJECT NO:
R5THS00138-JC

SHEET NO:
01 / 25

MASTER DRAWING INDEX - 23 INSTRUMENT HOOK-UP & LOOP DIAGRAM TEMPLATES

SHT	DRAWING NO	EQUIPMENT CATEGORY / TEMPLATE TITLE	REV	STATUS
03	KAL-JC-TYP-001	DUST / PARTICULATE ANALYZER HOOK-UP & LO.. Mfr: ENVEA / PCME Model: PCME QAL 991 (Electrodynamic	3.6	APPROVED
04	KAL-JC-TYP-002	COMBUSTIBLE LEL GAS DETECTOR HOOK-UP & L.. Mfr: Sensidyne / SmartGas Model: 925FGD Explosionproof Hydroc	3.6	APPROVED
05	KAL-JC-TYP-003	pH ANALYTICAL MEASUREMENT LOOP HOOK-UP &.. Mfr: Mettler Toledo Model: M300G2 4-Wire Transmitter +	3.6	APPROVED
06	KAL-JC-TYP-004	DIFFERENTIAL PRESSURE GAUGE HOOK-UP & ME.. Mfr: Ashcroft Model: Model 1132 (Piston-Type Diff	3.6	APPROVED
07	KAL-JC-TYP-005	DIFFERENTIAL PRESSURE TRANSMITTER HOOK-U.. Mfr: Endress+Hauser Model: Deltabar PMD55B (Piezoresist	3.6	APPROVED
08	KAL-JC-TYP-006	CORIOLIS MASS FLOWMETER HOOK-UP & LOOP D.. Mfr: Endress+Hauser Model: Promass E 300 (8E3B50 / Dual	3.6	APPROVED
09	KAL-JC-TYP-007	ELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP.. Mfr: Endress+Hauser Model: Promag P 300 (5P3B50 / 80 /	3.6	APPROVED
10	KAL-JC-TYP-008	VORTEX STEAM FLOWMETER HOOK-UP & LOOP DI.. Mfr: Endress+Hauser Model: Prowirl F 200 (7F2C80 / 1H w	3.6	APPROVED
11	KAL-JC-TYP-009	THERMAL MASS GAS FLOWMETER HOOK-UP & LOO.. Mfr: Endress+Hauser Model: t-mass F 300 / I 300 (6F3B /	3.6	APPROVED
12	KAL-JC-TYP-010	ULTRASONIC CLAMP-ON FLOWMETER HOOK-UP & .. Mfr: Keyence Model: FD-H Series (FD-H20 / FD-H32	3.6	APPROVED
13	KAL-JC-TYP-011	MANUAL HAND VALVE POSITION SWITCH HOOK-U.. Mfr: TeroFox / APL-HKC Model: TF-20DFS Ball Valve + APL-21	3.6	APPROVED
14	KAL-JC-TYP-012	LIQUID LEVEL SWITCH (VIBRATING FORK) HOO.. Mfr: Endress+Hauser Model: Liquiphant FTL51B (Tuning Fo	3.6	APPROVED
15	KAL-JC-TYP-013	SOLID LEVEL SWITCH (POWDER FORK) HOOK-UP.. Mfr: Endress+Hauser Model: Soliphant FTM51 (Vibrating F	3.6	APPROVED

SHT	DRAWING NO	EQUIPMENT CATEGORY / TEMPLATE TITLE	REV	STATUS
16	KAL-JC-TYP-014	RADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP.. Mfr: Endress+Hauser Model: Micropilot FMR67B (80 GHz Hi	3.6	APPROVED
17	KAL-JC-TYP-015	PRESSURE GAUGE & SIPHON HOOK-UP & MECHAN.. Mfr: Ashcroft Model: Model T5500 (100mm Dial All-	3.6	APPROVED
18	KAL-JC-TYP-016	ELECTRONIC PRESSURE SWITCH HOOK-UP & LOO.. Mfr: Endress+Hauser Model: Ceraphant PTC31B (Ceramic Se	3.6	APPROVED
19	KAL-JC-TYP-017	PRESSURE TRANSMITTER HOOK-UP & LOOP DIAG.. Mfr: Endress+Hauser Model: Cerabar PMP51B / PMC51B (Met	3.6	APPROVED
20	KAL-JC-TYP-018	BIMETAL TEMPERATURE GAUGE & THERMOWELL H.. Mfr: Ashcroft Model: Model FI (100mm Dial Everyan	3.6	APPROVED
21	KAL-JC-TYP-019	TEMPERATURE TRANSMITTER (RTD Pt100) HOOK.. Mfr: Endress+Hauser Model: iTEMP TMT71 + Omnigrad M TM4	3.6	APPROVED
22	KAL-JC-TYP-020	PNEUMATIC ON-OFF CONTROL VALVE HOOK-UP &.. Mfr: TeroFox / El-O-Matic Model: TF-20DFS Ball Valve + F-Seri	3.6	APPROVED
23	KAL-JC-TYP-021	MODULATING GLOBE CONTROL VALVE HOOK-UP &.. Mfr: Samson Model: Type 3241 Globe Valve + Trov	3.6	APPROVED
24	KAL-JC-TYP-022	ELECTRONIC TEMPERATURE SWITCH HOOK-UP & .. Mfr: Endress+Hauser Model: Thermophant TTR35 (Compact P	3.6	APPROVED
25	KAL-JC-TYP-023	PNEUMATIC TURBINE VIBRATOR HOOK-UP & LOO.. Mfr: Netter Vibration Model: NCT 5 Pneumatic Turbine Vibr	3.6	APPROVED

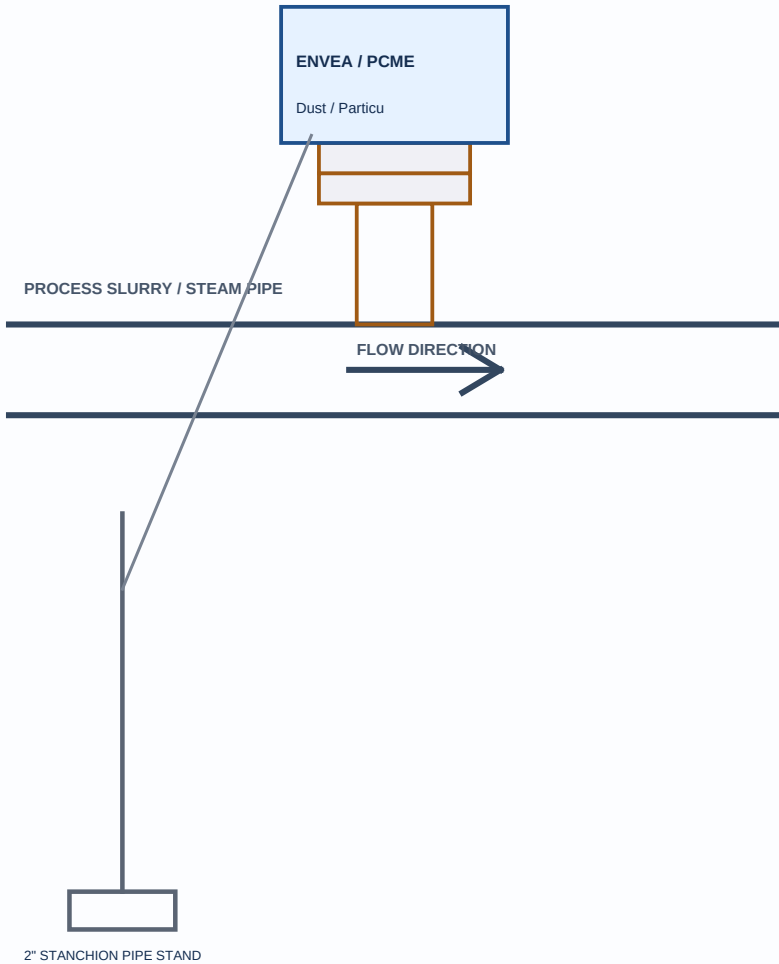
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
INSTRUMENT TEMPLATE DRAWING DIRECTORY INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM		DWG NO: KAL-JC-TYP-000-IDX REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 02 / 25

TEMPLATE TYP-01: DUST / PARTICULATE ANALYZER

TYPICAL TAGS: AT-60241 Series

SERVICE: Exhaust Duct Dust Emission Monitoring | PLC I/O: 1756-IF16 (AI) & 1756-IB32 (DI Alarm)

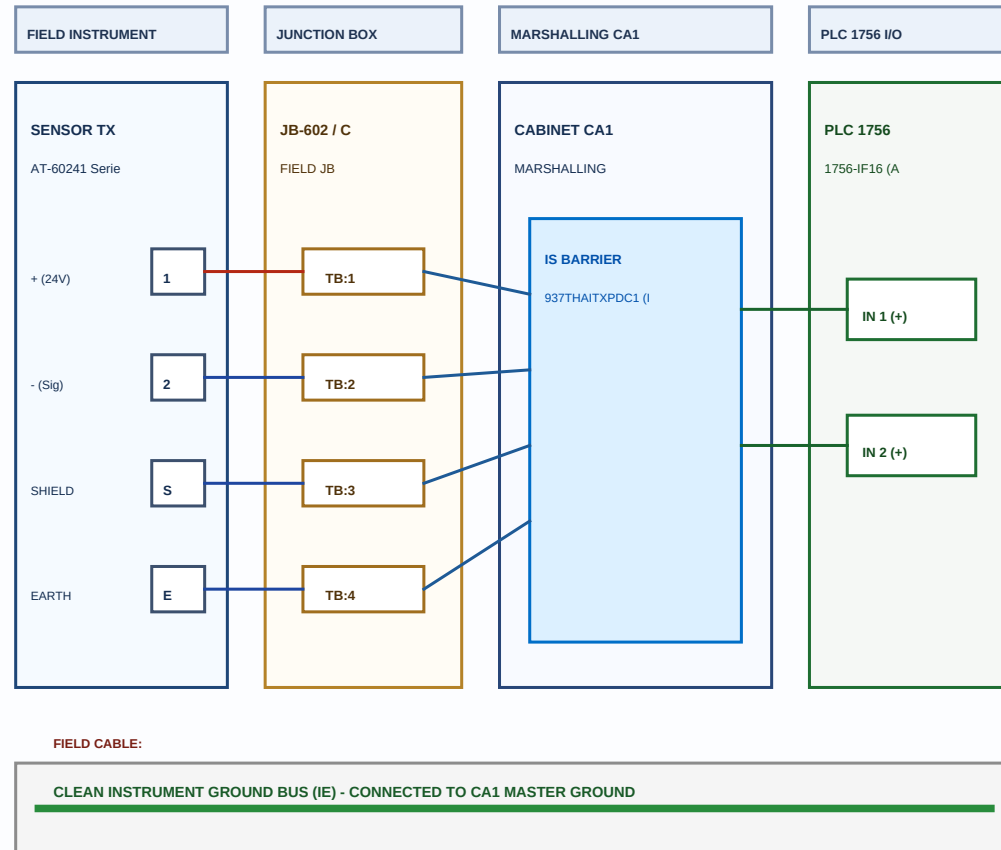
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN100 Flange w/ Purge Air (1..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 Weatherproof Cast Alumi..
Hazardous Class:	ATEX Zone 21/22 Dust Ex tb I..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 4-20mA Dust Conc / DI: Fault Contact
Power Supply:	24V DC / 15W + Clean Dry Instrument Air
Field Cable Type:	2Px1.5mm ² IS Shielded Overall (Belden ..)
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Analog Barrier)
PLC Module:	1756-IF16 (AI) & 1756-IB32 (DI Alarm)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENVEA / PCME

MODEL: PCME QAL 991 (Electrodynamic / Optical)

SERVICE: Exhaust Duct Dust Emission Monitoring

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Dust Transmitter Sensor Unit PCME QAL 991		1 EA
2	Mating Counter-Flange DN100 PN Carbon Steel / SS316		1 EA
3	Purge Air Filter Regulator & G SMC 1/4" BSP w/ Bracket		1 SET
4	Cable Gland M20x1.5 Ex d/e Nickel-plated Brass		2 EA
5	Instrument Stand 2" Pipe & Bas Hot-dip Galvanized Steel		1 SET

ENGINEERING INSTALLATION NOTES

1. Mount sensor on straight run min. 5D downstream, 2D upstream of duct bends.
2. Instrument air purge must remain continuously ON to prevent window fouling.
3. Shield drain wire must be isolated in field and grounded only at Marshalling Cabinet CA1 earth bar.

INGREDION (THAILAND) CO., LTD.

KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING

SYSTEMS INTEGRATION & AUTOMATION

DUST / PARTICULATE ANALYZER HOOK-UP & LOOP DIAGRAM

INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DWG NO: KAL-JC-TYP-001

REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG

CHECKED: LEAD-PE

STATUS: APPROVED

SCALE: N.T.S.

PROJECT NO:

R5THS00138-JC

SHEET NO:

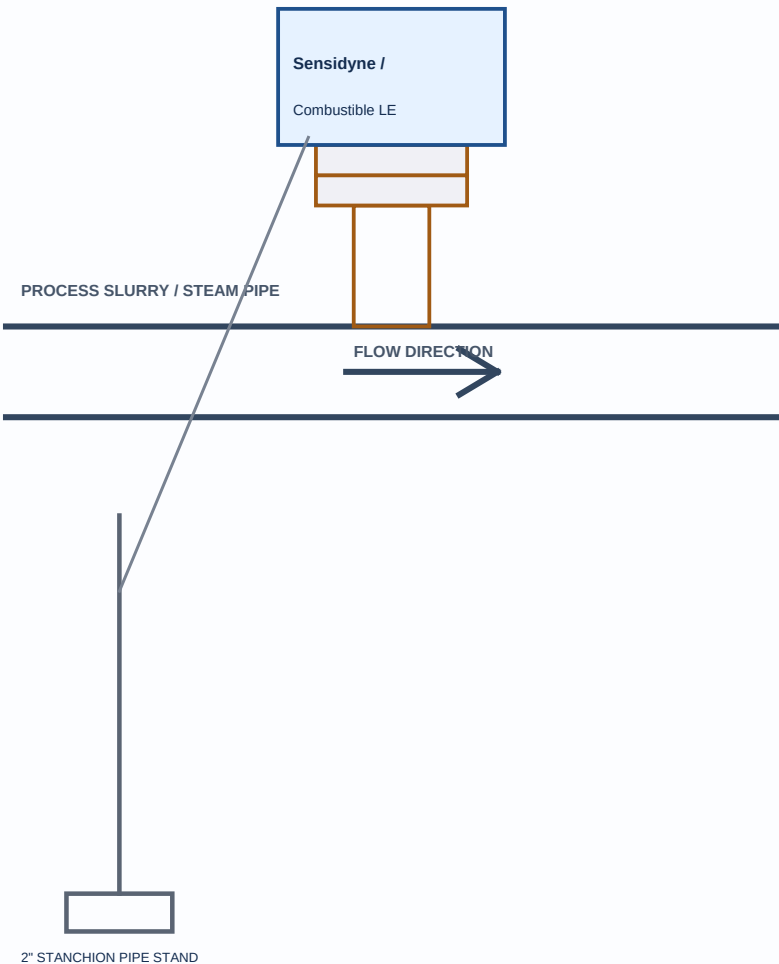
03 / 25

TEMPLATE TYP-02: COMBUSTIBLE LEL GAS ANALYZER

TYPICAL TAGS: AIT-60101 / AIT-60201

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor | PLC I/O: 1756-IF16 (AI LEL) & 1756-IB32 (DI Alarm)

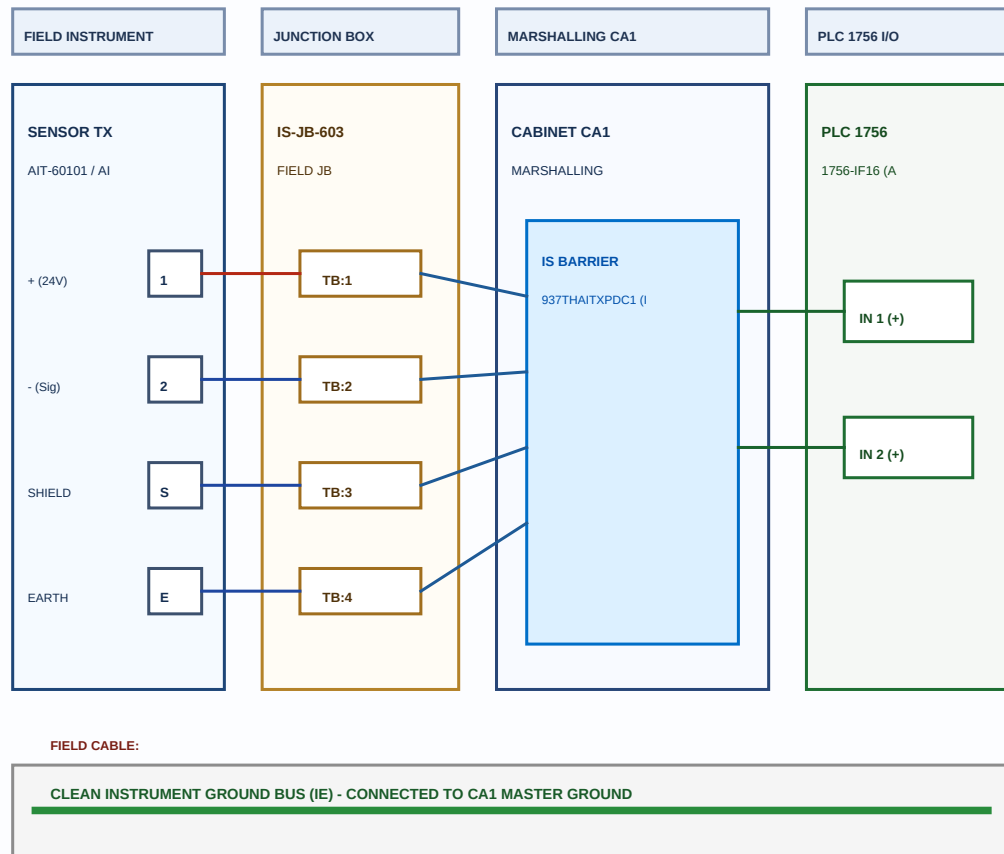
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Ambient Air Diffusion / Spla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66 Heavy-duty Ex..
Hazardous Class:	Class I, Div 1, Grp BCD / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-100% LEL (4-20mA) / DI: Gas Alarm
Power Supply:	24V DC (18-32V DC, 200mA max)
Field Cable Type:	3Cx1.5mm² Shielded Fire-Resistant (Bel..
Destination JB:	IS-JB-603 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (AI LEL) & 1756-IB32 (DI Ala..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SENSIDYNE / SMARTGAS

MODEL: 925FGD Explosionproof Hydrocarbon Gas Detector

SERVICE: Explosion Hazard Lower Explosive Limit (LEL) Monitor

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Gas Detector Transmitter 925FG Sensidyne Infrared / Catalytic		1 EA
2	Weather Guard / Dust Protectio Hydrophobic SS316		1 EA
3	Calibration Adapter & Cup Molded PVC / Viton O-ring		1 EA
4	Explosionproof Cable Gland 3/4 SS316 Flameproof Ex d		1 EA
5	Wall / Stanchion Mounting Brac SS304 Universal Bracket		1 SET

ENGINEERING INSTALLATION NOTES

- Install 0.3m to 0.5m above floor for gases heavier than air, or near ceiling for lighter gases.
- Field sensor must undergo zero and span gas calibration prior to plant commissioning.
- Wiring must run in rigid galvanized steel conduit or approved armored flameproof cable.

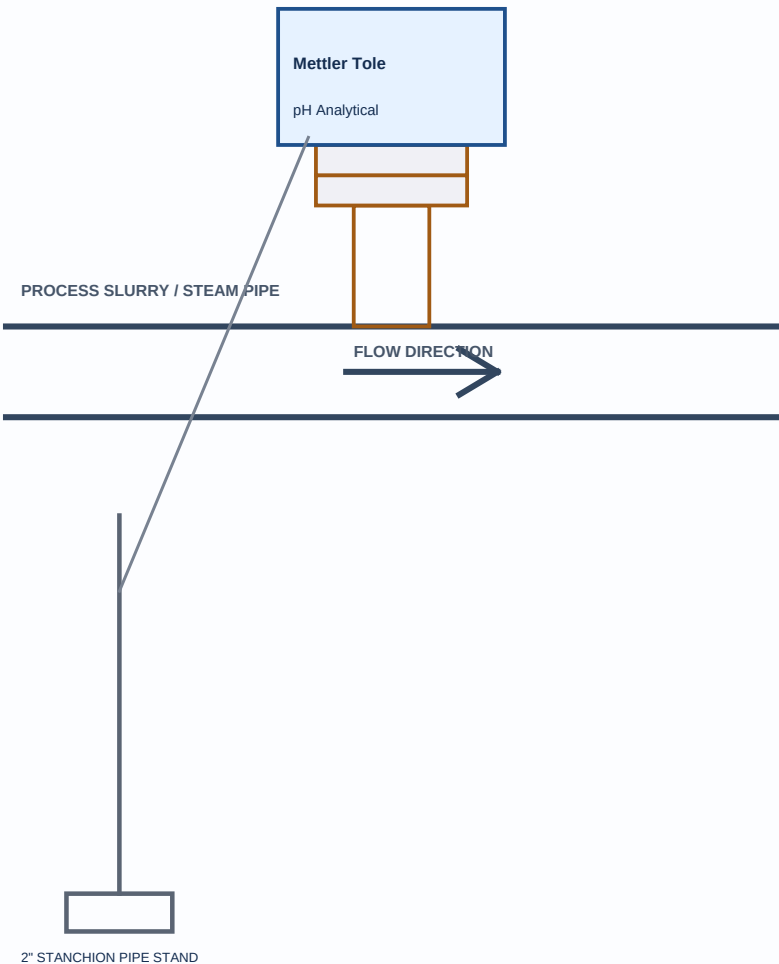
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCOMBUSTIBLE LEL GAS DETECTOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-002
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
04 / 25

TEMPLATE TYP-03: PH ANALYTICAL LOOP

TYPICAL TAGS: AIT-40101 / AIT-40201

SERVICE: Cooker Slurry & Flash Tank pH Neutralization | PLC I/O: 1756-IF16 (AI pH) & 1756-IB32 (DI Alarms)

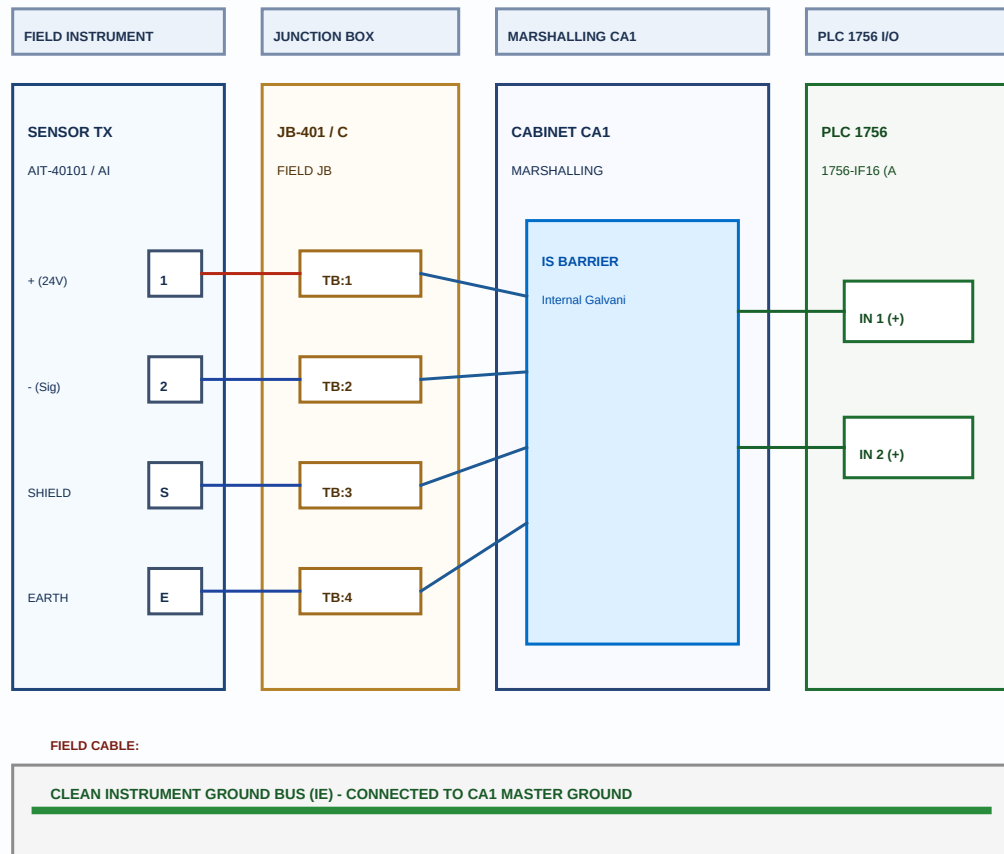
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN25 Weld-in Socket (Incline..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP65 Polycarbonate..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6/T4..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-14 pH (4-20mA HART) / AI: Temp
Power Supply:	24V DC / 10W (Optional 100-240V AC)
Field Cable Type:	VP6 High-Z Cable to Tx + 2Px1.5mm² Shi..
Destination JB:	JB-401 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation (4-Wire Tx)
PLC Module:	1756-IF16 (AI pH) & 1756-IB32 (DI Alar..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: METTLER TOLEDO

MODEL: M300G2 4-Wire Transmitter + InPro3250i + InTrac 777P

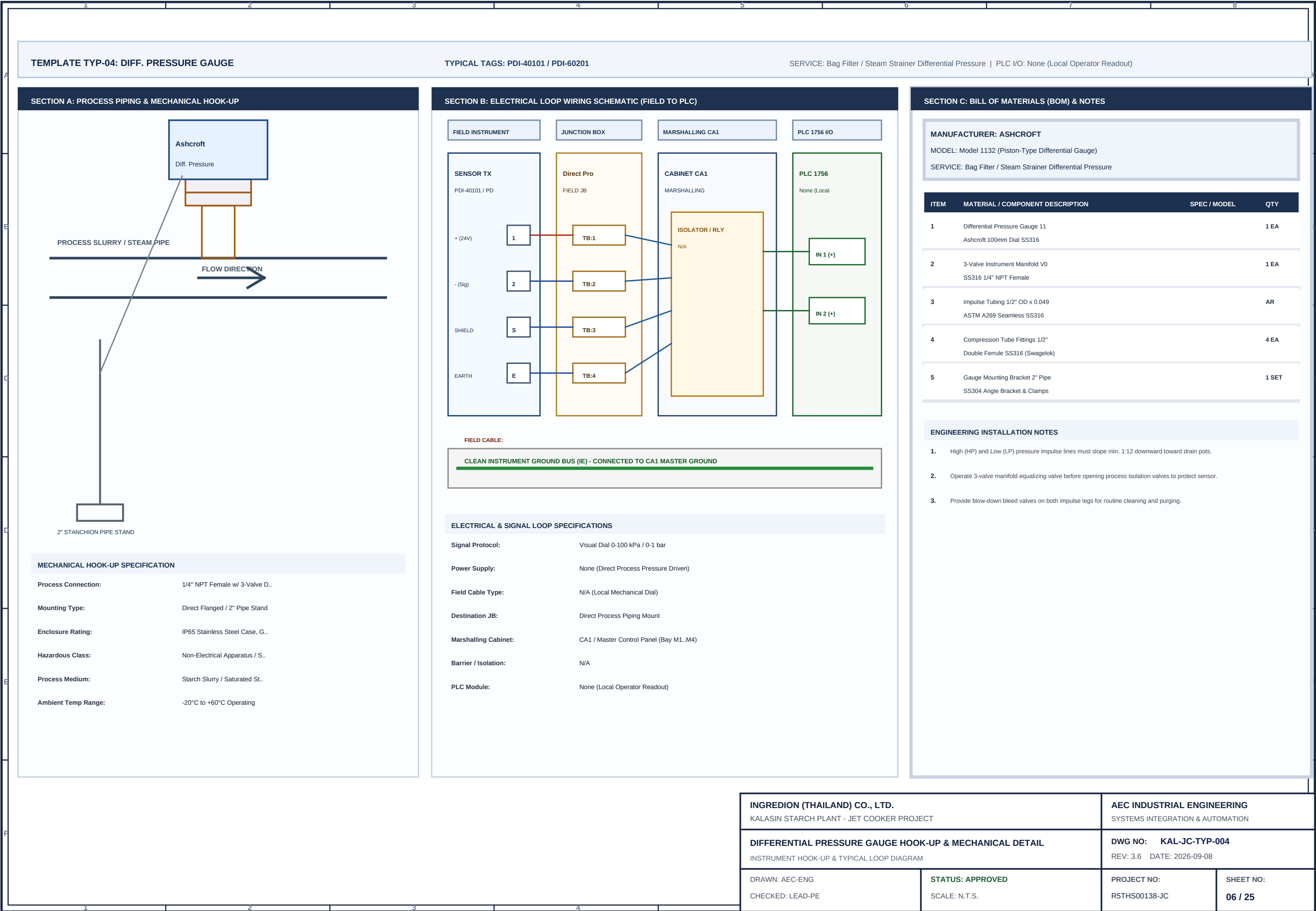
SERVICE: Cooker Slurry & Flash Tank pH Neutralization

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	pH Multi-Parameter Transmitter Mettler Toledo M300 4-Wire		1 EA
2	Intelligent pH Sensor InPro325 Pre-pressurized Liquid Gel		1 EA
3	Retractable Housing InTrac 777 SS316L / Hastelloy C w/ Flange		1 EA
4	Weld-in Socket 40mm / DN25 (15 SS316L Sloped Socket		1 EA
5	2" Pipe Mounting Kit for M300G SS304 Bracket & U-Bolts		1 SET

ENGINEERING INSTALLATION NOTES

- Weld socket must be installed inclined at min. 15° above horizontal to prevent air bubble entrapment.
- Use retractable housing safety interlock valve before extracting probe during process operation.
- Ground transmitter chassis directly to plant PE bar with min. 4mm² copper grounding strap.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTpH ANALYTICAL MEASUREMENT LOOP HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-003
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
05 / 25

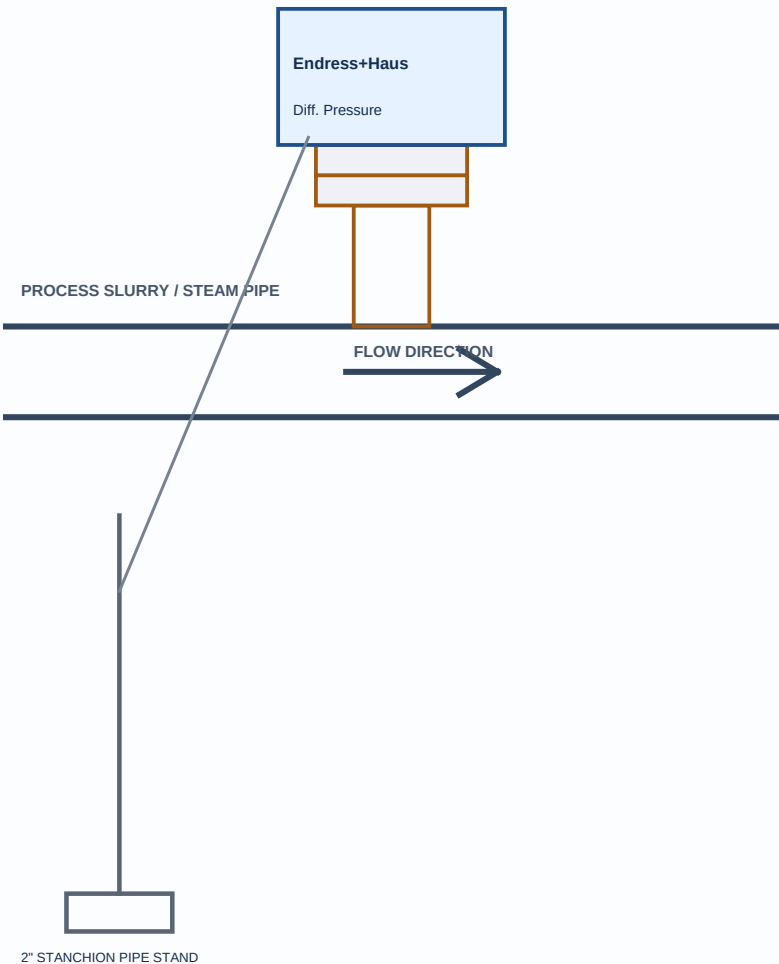


TEMPLATE TYP-05: DIFF. PRESSURE TRANSMITTER

TYPICAL TAGS: DPT-60201 / DPT-60901

SERVICE: Filter Differential Pressure / Orifice Flow Drop | PLC I/O: 1756-IF16 (16-Point Diff Analog Input)

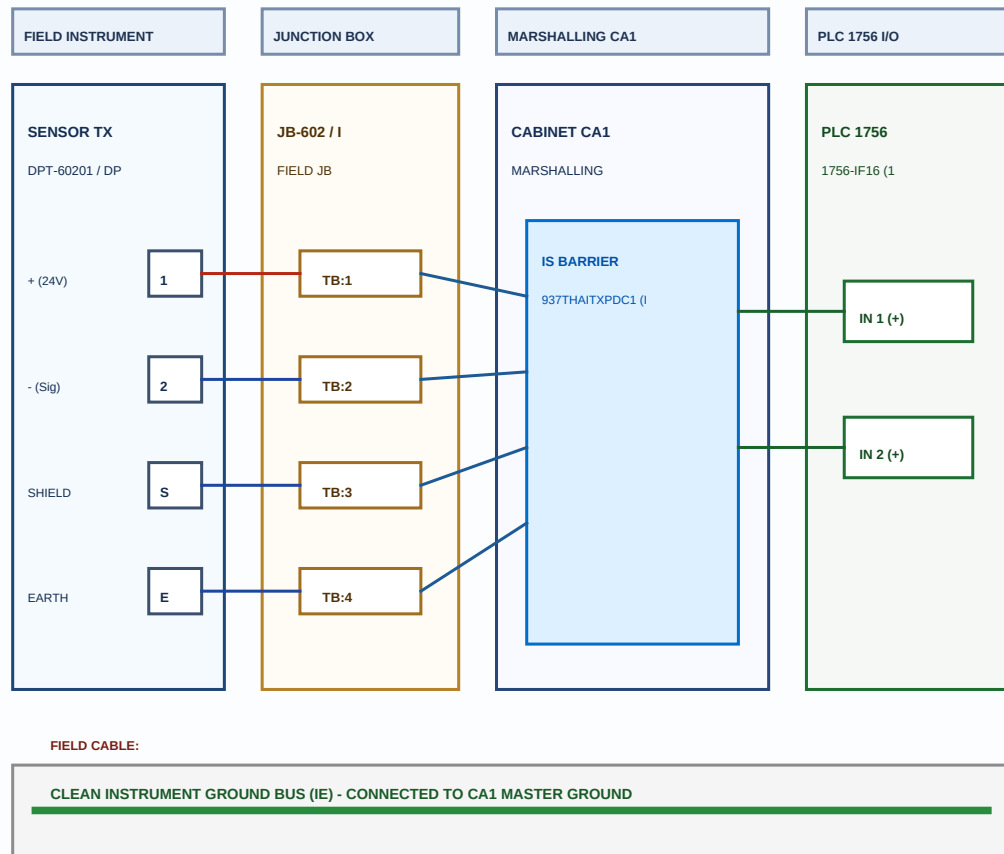
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/4" NPT Female w/ 5-Valve M..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual-Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Process Delta-P (0-500 mbar)
Power Supply:	24V DC via 1756-IF16 Analog Input Loop
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 876..
Destination JB:	JB-602 / IS-JB-603
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Diff Analog Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Deltabar PMD55B (Piezoresistive / Ceramic)

SERVICE: Filter Differential Pressure / Orifice Flow Drop

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Deltabar PMD55B Transmitter E+H Ceramic / SS316 Diaphragm		1 EA
2	5-Valve Manifold DA63M (Flange SS316 Integrated Bleed Valves		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Stanchion Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-Plated Brass Armored Glan		1 EA
5	Impulse Tubing 1/2" OD x 0.049 ASTM A269 Seamless SS316		AR

ENGINEERING INSTALLATION NOTES

- Mount transmitter below process tapping points for liquid service to maintain filled impulse lines.
- 5-valve manifold sequence: Open equalizing, open HP & LP blocks, close equalizing, check zero.
- Connect 2-wire loop to PLC 1756-IF16 terminals: (+) to 24V loop power, (-) to Channel Input.

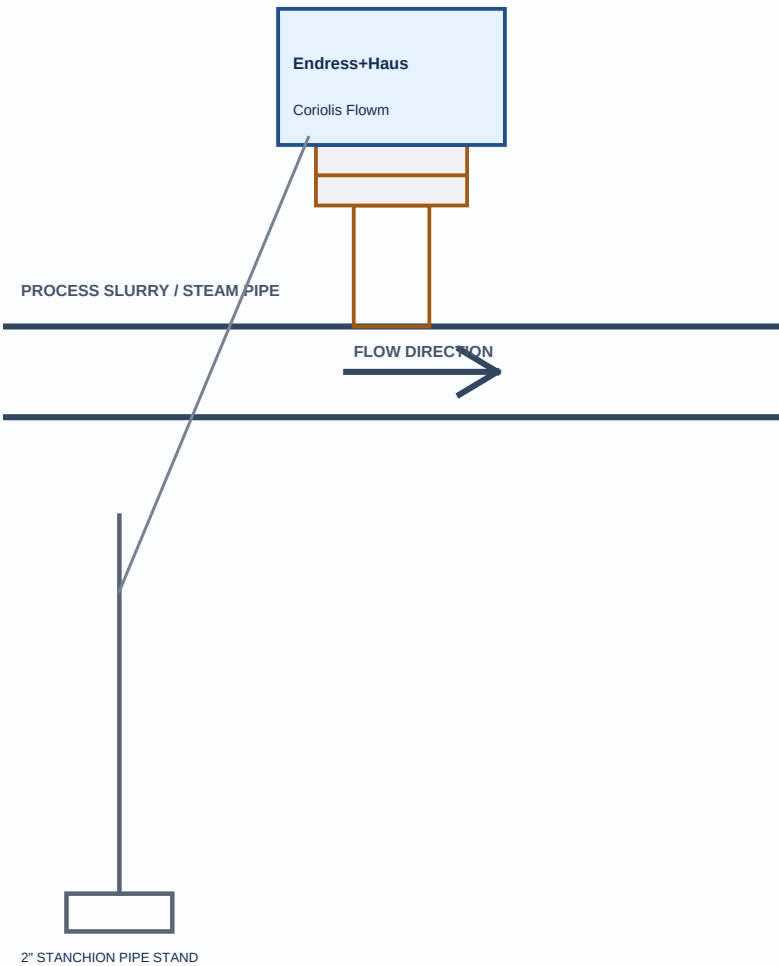
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTDIFFERENTIAL PRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-005
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
07 / 25

TEMPLATE TYP-06: CORIOLIS FLOWMETER

TYPICAL TAGS: FT-40201 Series

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density | PLC I/O: 1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Density)

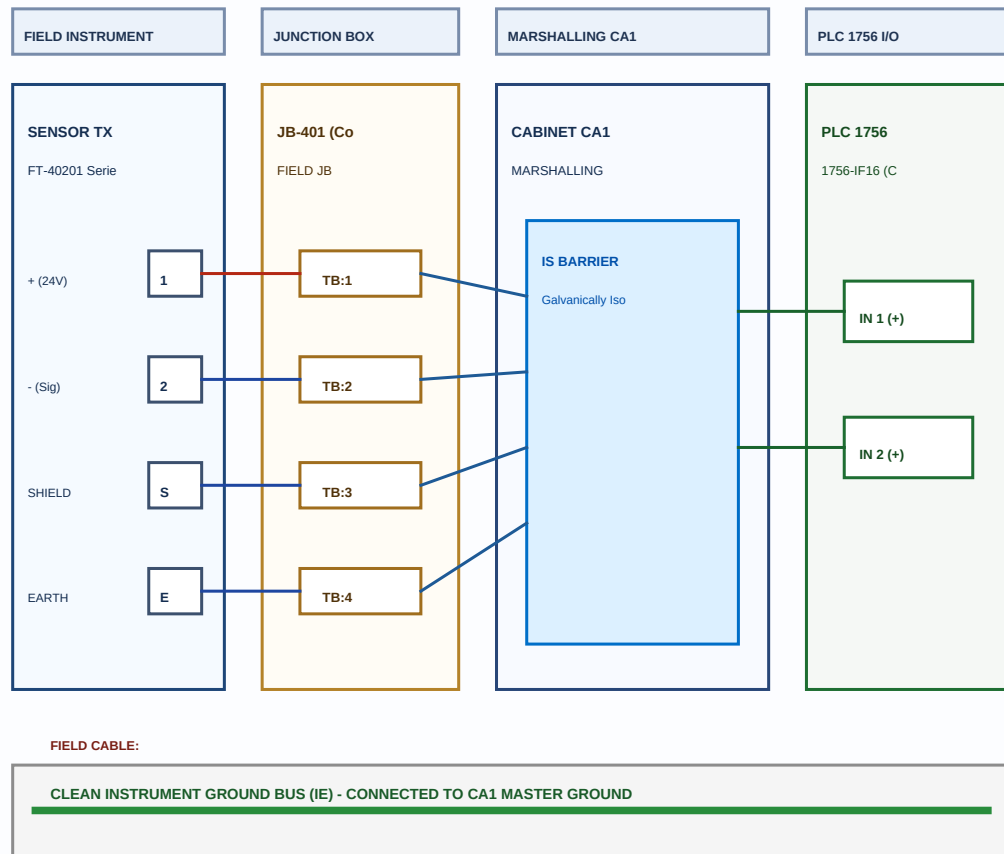
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50 (2") ASME B16.5 Cl.150 ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP67 Die-cast Aluminum Trans..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Mass (kg/h) & Density (g/c..
Power Supply:	24V DC / 10W (Transmitter Electronics)
Field Cable Type:	2Px1.5mm² Shielded Overall (Belden 877..
Destination JB:	JB-401 (Coker Section 1)
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanically Isolated Active 4-20mA Ou..
PLC Module:	1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Den..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promass E 300 (8E3B50 / Dual Tube SS316L)

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promass E 300 Coriolis Flowmet E+H DN50 Cl.150 SS316L Dual Tube		1 EA
2	Companion Counter-Flanges DN50 ASTM A182 F316 / RF Flanges		2 EA
3	PTFE / Spiral Wound Gaskets 316SS with PTFE Filler		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		2 EA
5	Pipe Support Clamps & Structur Heavy Duty Rigidity Anchors		2 SET

ENGINEERING INSTALLATION NOTES

- Sensor tubes must be installed pointing downwards (flag-down) for liquid slurry to prevent gas collection.
- Support pipeline rigidly on both upstream and downstream sides within 2 pipe diameters of meter flanges.
- Do not clamp or attach pipe supports directly to the Coriolis sensor housing.

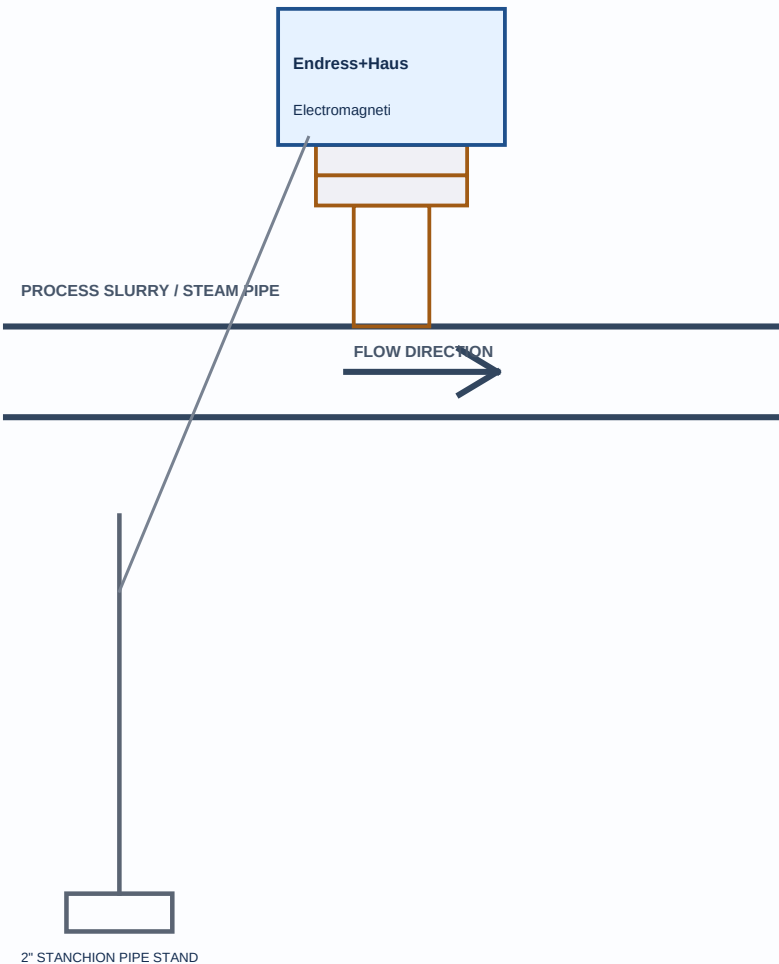
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCORIOLIS MASS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-006
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
08 / 25

TEMPLATE TYP-07: ELECTROMAGNETIC FLOWMETER

TYPICAL TAGS: FT-40202 / FT-60201

SERVICE: Slurry & Process Water Volumetric Flow Measurement | PLC I/O: 1756-IF16 (4-20mA Flow) & 1756-IB32 (Pulse Counter)

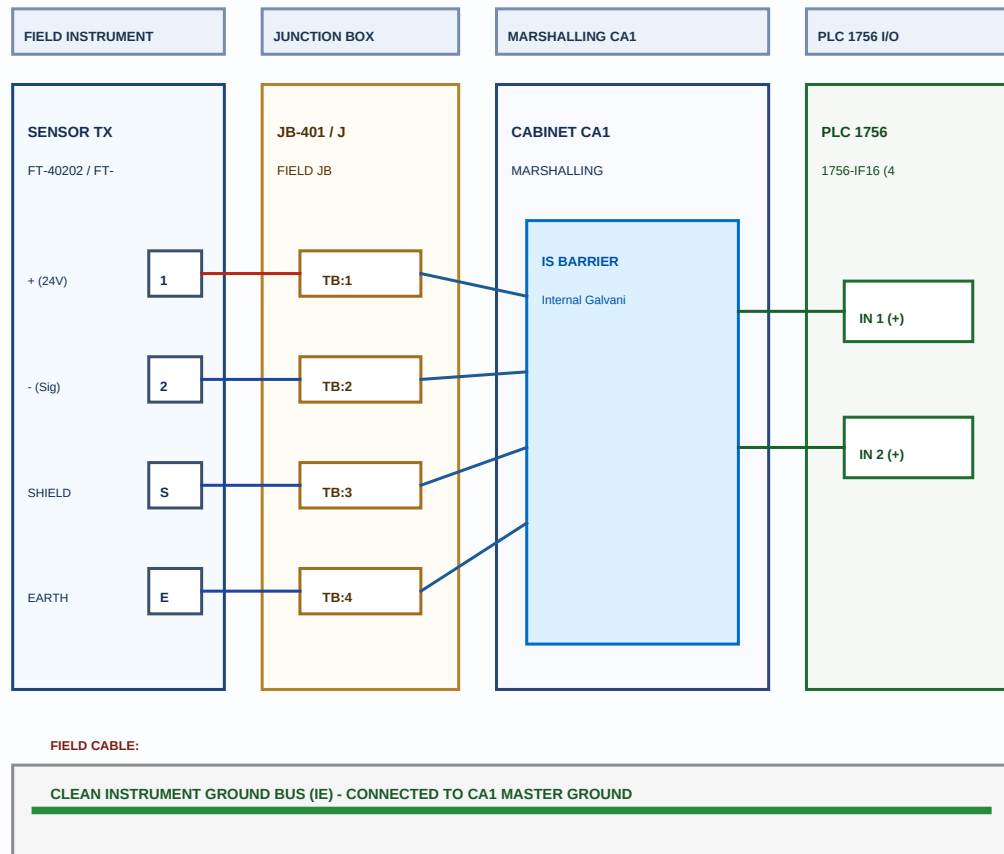
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50/80/100 ASME CL150 RF F..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Tran..
Hazardous Class:	ATEX II 2G Ex eb Iia IIC T6....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



FIELD CABLE:

CLEAN INSTRUMENT GROUND BUS (IE) - CONNECTED TO CA1 MASTER GROUND

ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-150 m³/h) + Scaled Pulses
Power Supply:	24V DC / 8W (4-Wire Electronics)
Field Cable Type:	1Px1.5mm² (Analog) + 1Px1.5mm² (Pulse/..
Destination JB:	JB-401 / JB-601
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation
PLC Module:	1756-IF16 (4-20mA Flow) & 1756-IB32 (P..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promag P 300 (5P3B50 / 80 / 1H PTFE Liner)

SERVICE: Slurry & Process Water Volumetric Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promag P 300 Magmeter Sensor & E+H PTFE Lined, Hastelloy C Elec		1 EA
2	Grounding Rings / Discs SS316L E+H Grounding Electrodes Kit		2 EA
3	PTFE Envelope Flange Gaskets Chemical Resistant Gaskets		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Glands		2 EA
5	Grounding Cables 6mm² Green/Ye Copper Earthing Strap with Lugs		2 EA

ENGINEERING INSTALLATION NOTES

- Straight piping requirements: min. 5D upstream and 2D downstream of flowmeter.
- Install grounding rings between meter flanges and pipe counter-flanges to ensure reference ground.
- Pipe must always remain 100% full of conductive liquid for accurate flow measurement.

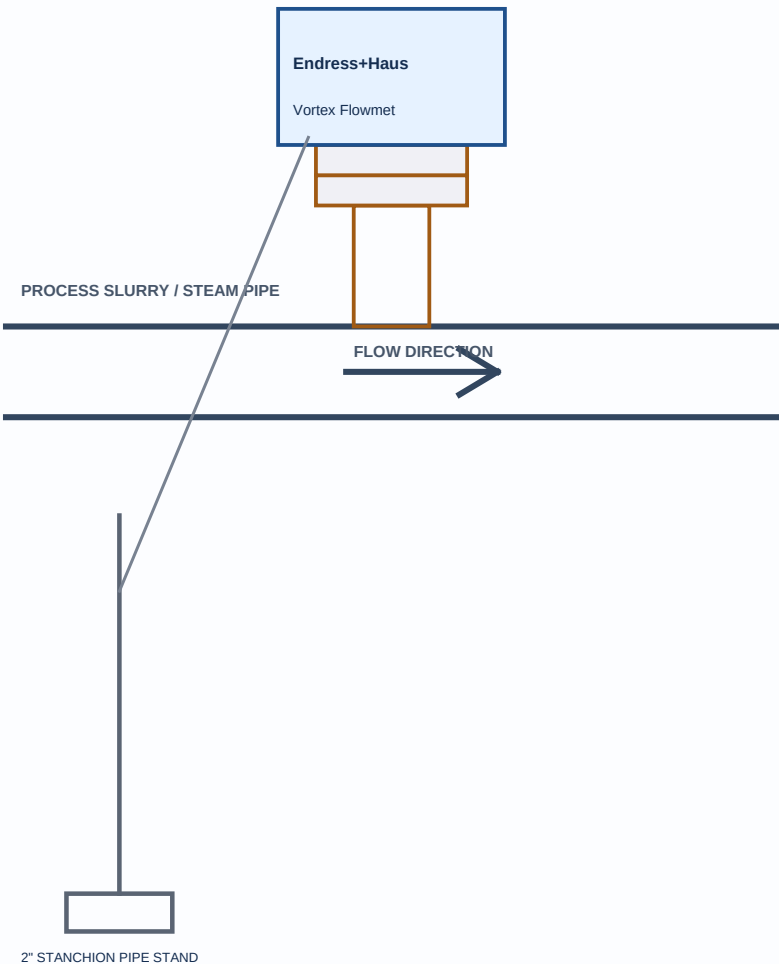
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-007
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
09 / 25

TEMPLATE TYP-08: VORTEX FLOWMETER

TYPICAL TAGS: FT-60101 / FT-60202

SERVICE: High-Pressure Saturated Steam Flow Measurement | PLC I/O: 1756-IF16 (Compensated Mass Flow Steam)

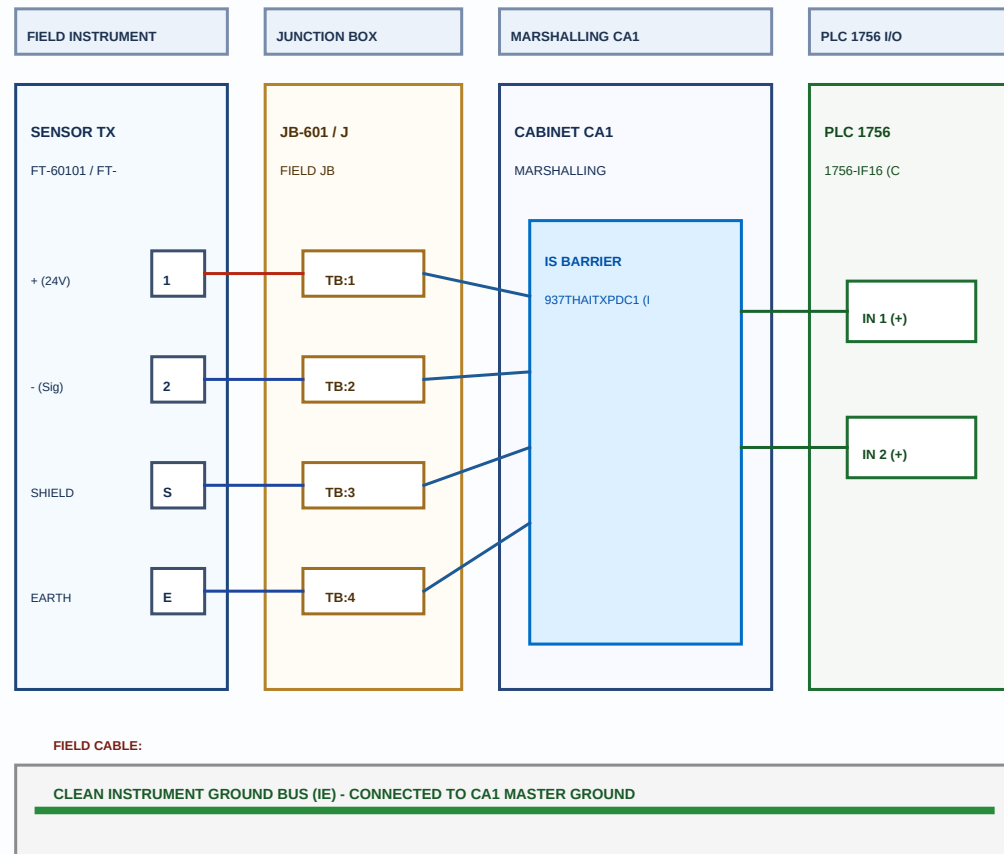
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80/100 (3"/4") ASME Cl.300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Dual-Compartment IP67 Die-ca..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Steam Flow (0-10,000 kg/h)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (High Temp ..
Destination JB:	JB-601 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Loop Powered Repeater)
PLC Module:	1756-IF16 (Compensated Mass Flow Steam)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Prowirl F 200 (7F2C80 / 1H with Temp Sensor)

SERVICE: High-Pressure Saturated Steam Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Prowirl F 200 Flanged Vortex M E+H SS316 Cast Body w/ DSC Senso		1 EA
2	Counter-Flanges DN80 Cl.300 RF ASTM A105 / A182 F316 Flanges		2 EA
3	Spiral Wound Gaskets Cl.300 Gr Graphite Filled SS316 Rings		2 EA
4	Cable Gland M20x1.5 Ex d/e Hig Brass Nickel-Plated Gland		1 EA
5	Pipeline Insulation Spacer & S Thermal Extension Neck Jacket		1 SET

ENGINEERING INSTALLATION NOTES

- Straight piping: min. 15D upstream and 5D downstream from flow disturbers (valves, reducers).
- Insulate pipe line up to meter neck, but leave transmitter electronics housing uninsulated.
- Verify that internal integrated Pt100 temperature compensation is activated in HART configuration.

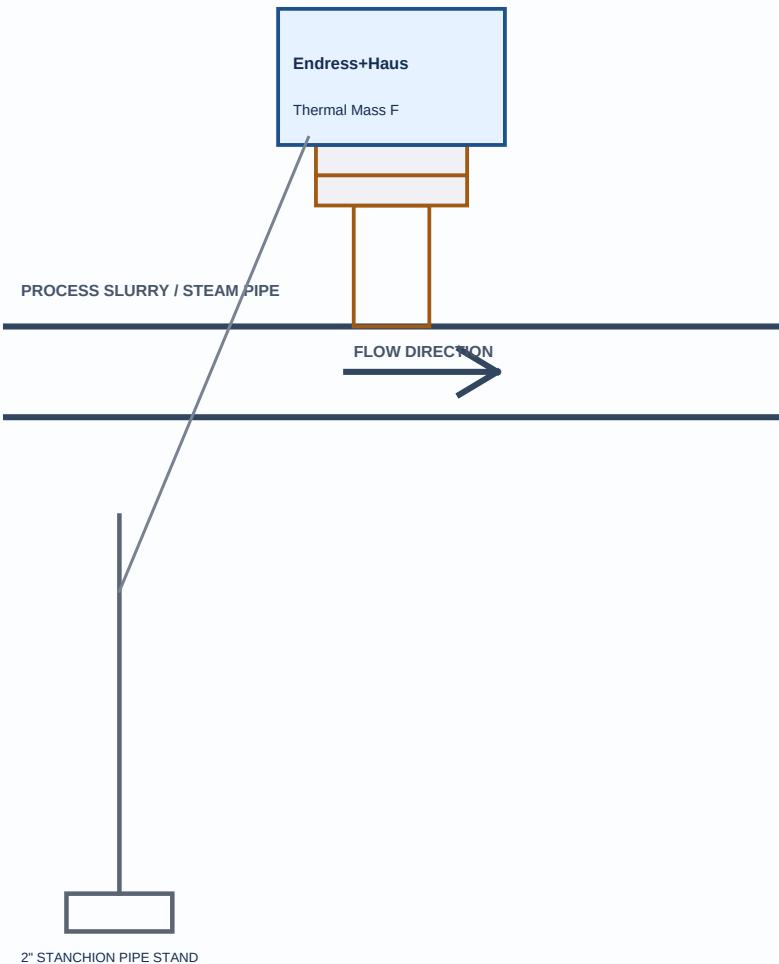
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONVORTEX STEAM FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-008
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
10 / 25

TEMPLATE TYP-09: THERMAL MASS FLOWMETER

TYPICAL TAGS: FT-80301 Series

SERVICE: LPG / Natural Gas Boiler Fuel Consumption | PLC I/O: 1756-IF16 (AI Gas Flow) & 1756-IB32 (DI Alarm)

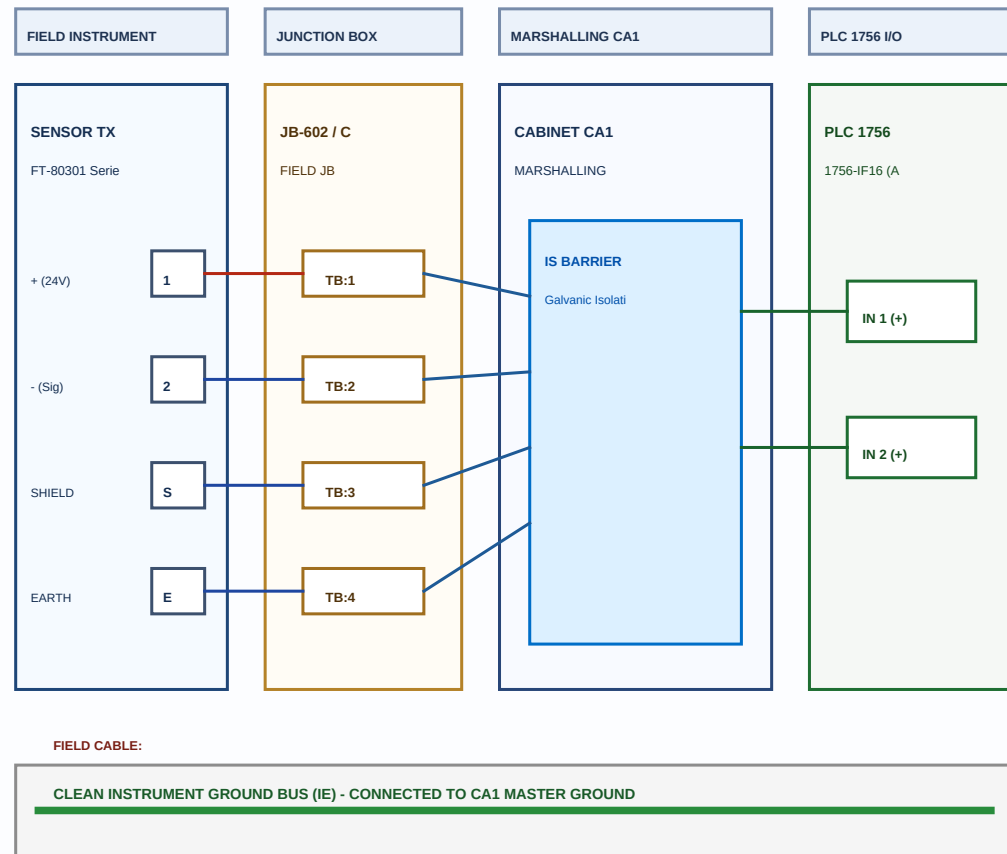
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN100 Flange / 1" NPT Retract..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Encl..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Standard Nm³/h Flow
Power Supply:	24V DC / 12W (4-Wire Electronics)
Field Cable Type:	2Px1.5mm² Shielded Armored (Belden 8777)
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanic Isolation / Ex d Installation
PLC Module:	1756-IF16 (AI Gas Flow) & 1756-IB32 (D..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: t-mass F 300 / I 300 (6F3B / 6I3B Insertion)

SERVICE: LPG / Natural Gas Boiler Fuel Consumption

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	t-mass 300 Thermal Flowmeter T E+H Insertion Probe w/ Hot-Tap		1 EA
2	Weld-in Boss 1" NPT Female Thr ASTM A105 / Carbon Steel Boss		1 EA
3	Isolation Ball Valve 1" Full B SS316 Cl.300 Gas Ball Valve		1 EA
4	Cable Gland M20x1.5 Ex d Flame Brass Gland for Armored Cable		2 EA
5	Grounding Cable 6mm² w/ Lug Copper Earthing Conductor		1 EA

ENGINEERING INSTALLATION NOTES

- Install probe at centerline of pipe (0.5 x Inside Diameter) matching flow direction arrow exactly.
- Verify straight pipe run: min. 20D upstream and 5D downstream of insertion tap.
- Equipotential bonding must be connected between meter body and gas piping system.

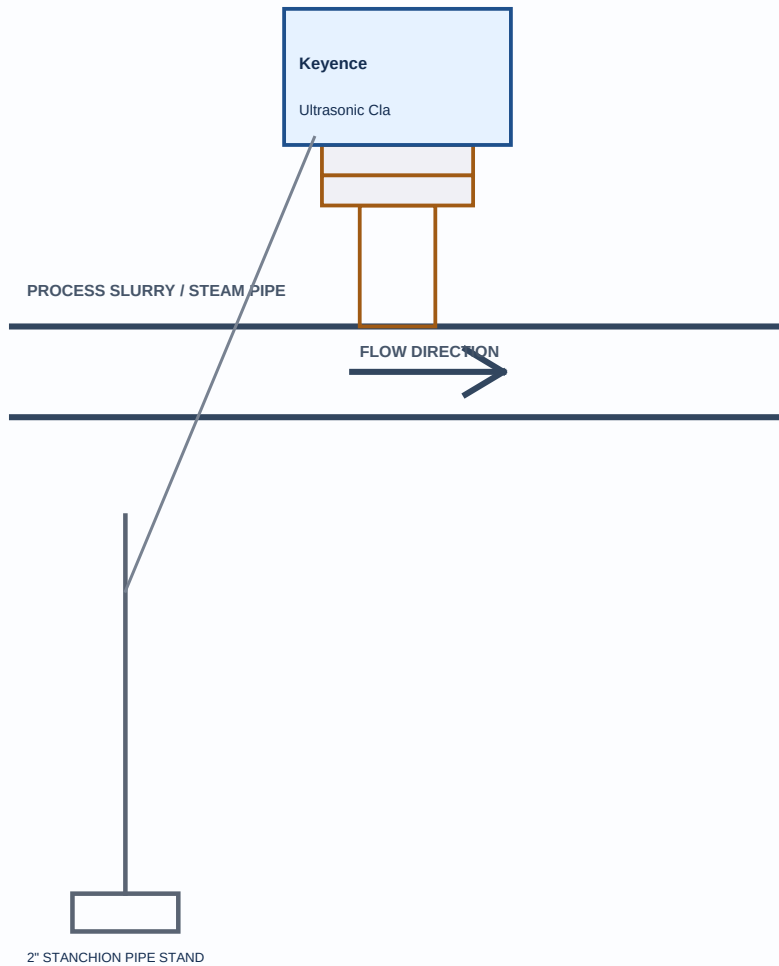
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONTHERMAL MASS GAS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-009
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
11 / 25

TEMPLATE TYP-10: ULTRASONIC CLAMP-ON FLOWMETER

TYPICAL TAGS: FIT-40105 Series

SERVICE: Cooling Water & Non-Invasive Slurry Verification | PLC I/O: 1756-IF16 (AI Flow Rate) & 1756-IB32 (Pulse)

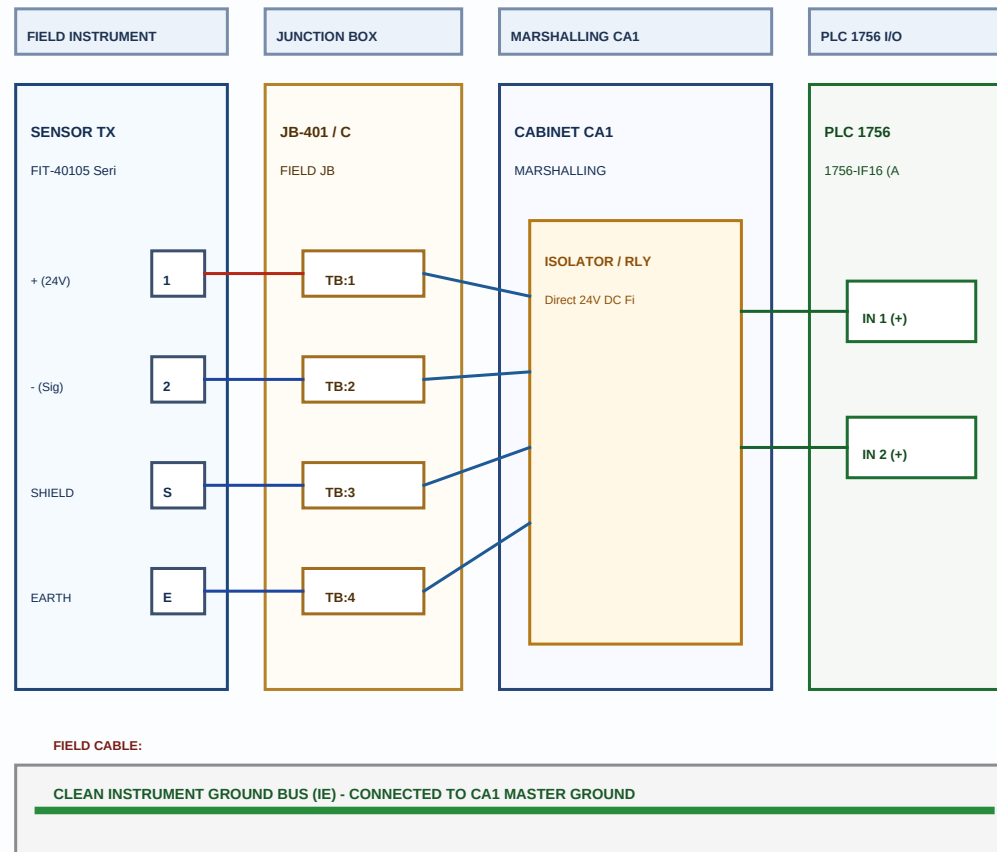
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	External Pipe Surface Mount ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP67 / IP69K Hygienic Stainl..
Hazardous Class:	General Industrial Area / No..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA (0-100 L/min) + NPN/PNP Pulse
Power Supply:	20-30V DC / 5W
Field Cable Type:	4Cx0.75mm² Factory Pre-Wired M12 Conne..
Destination JB:	JB-401 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Direct 24V DC Field Power
PLC Module:	1756-IF16 (AI Flow Rate) & 1756-IB32 (.

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: KEYENCE

MODEL: FD-H Series (FD-H20 / FD-H32 Clamp-on)

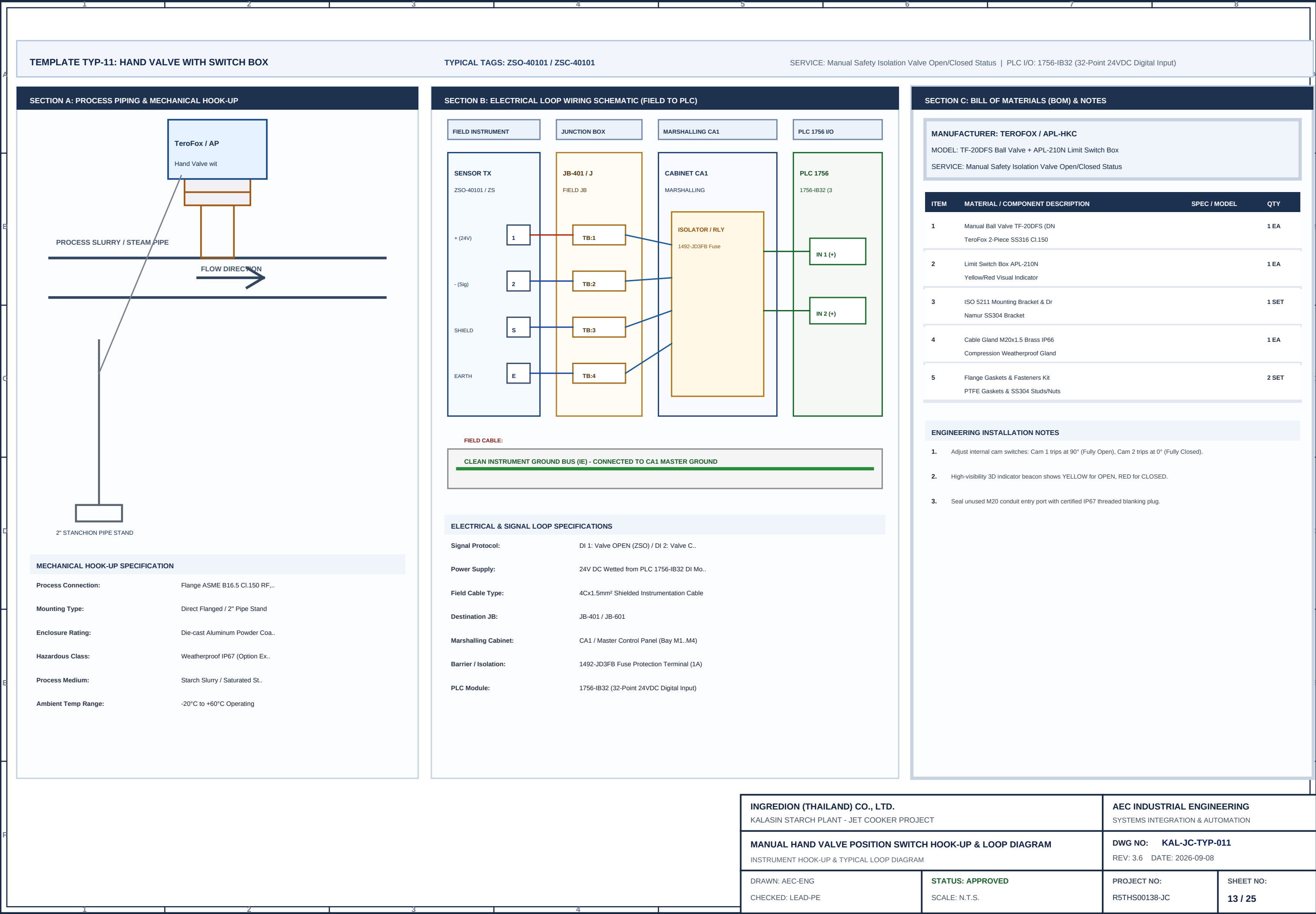
SERVICE: Cooling Water & Non-Invasive Slurry Verification

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Clamp-on Ultrasonic Sensor Hea Keyence FD-H20 / H32 Unit		1 EA
2	Acoustic Coupling Gel / Pad High-Stability Silicone Couplant		1 TUBE
3	Pipe Mounting Clamp Bands SS30 Adjustable Worm Gear Straps		2 EA
4	M12 4-Pin Shielded Cable (5m) PUR Oil-Resistant Flex Cable		1 EA
5	Field Termination Box (Local) IP66 Junction Box w/ DIN Rail		1 EA

ENGINEERING INSTALLATION NOTES

1. Clean pipe surface thoroughly with emery cloth to remove rust, paint, and scale before mounting.
2. Apply generous bead of acoustic coupling compound between sensor transducers and pipe wall.
3. Clamp unit securely on horizontal pipe at 3 o'clock or 9 o'clock position (never top or bottom).

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
ULTRASONIC CLAMP-ON FLOWMETER HOOK-UP & LOOP DIAGRAM INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM		DWG NO: KAL-JC-TYP-010 REV: 3.6 DATE: 2026-09-08	
DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED SCALE: N.T.S.	PROJECT NO: R5THS00138-JC	SHEET NO: 12 / 25

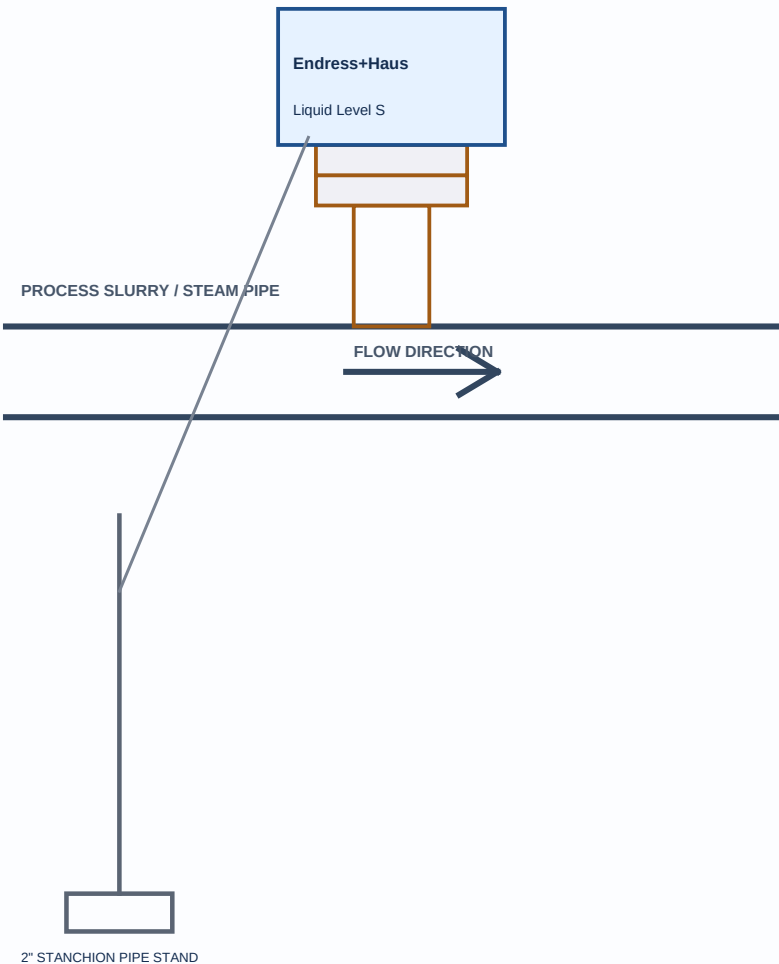


TEMPLATE TYP-12: LIQUID LEVEL SWITCH

TYPICAL TAGS: LSH-40101 / LSL-40101

SERVICE: High / Low Slurry Level Interlock & Pump Protection | PLC I/O: 1756-IB32 (24V DC Sinking/Sourcing Digital Input)

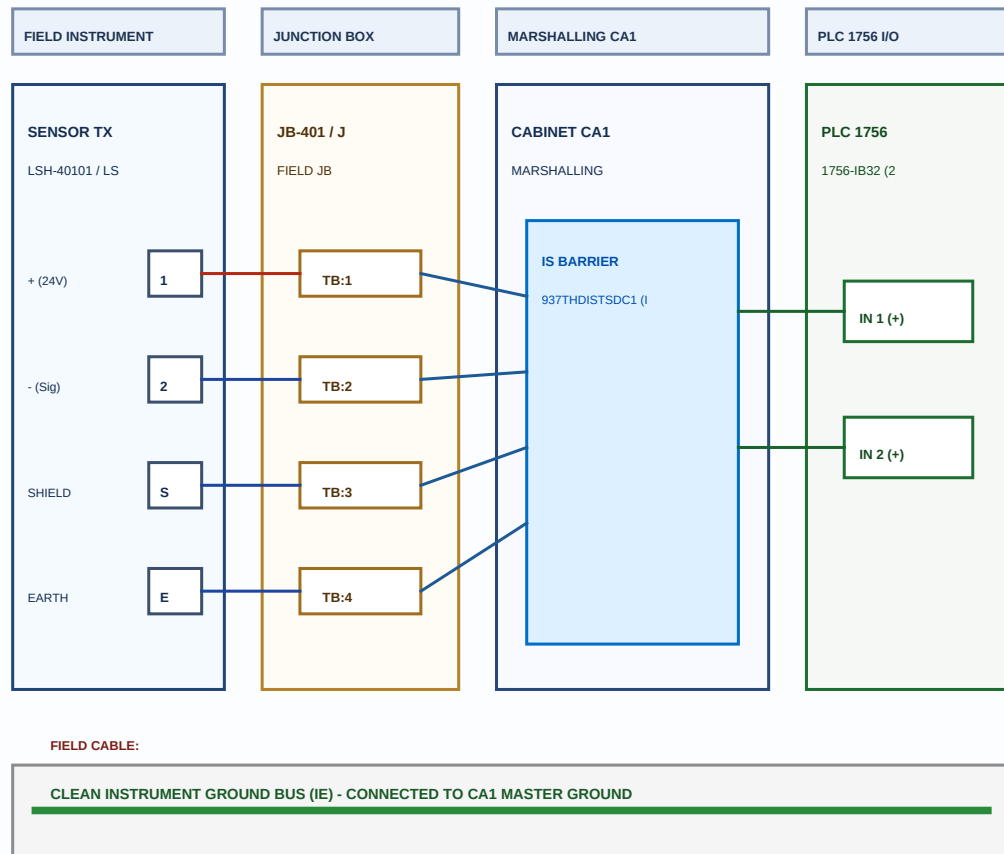
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	3/4" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Stainless ..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6.....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Level Alarm (Dry / Submerged ..
Power Supply:	24V DC (10-55V DC, PNP Mode)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THDISTSDC1 (IS Digital Input Barrier)
PLC Module:	1756-IB32 (24V DC Sinking/Sourcing Dig..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER
MODEL: Liquiphant FTL51B (Tuning Fork SS316L)
SERVICE: High / Low Slurry Level Interlock & Pump Protection

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Liquiphant FTL51B Level Switch E+H Tuning Fork SS316L (100mm)		1 EA
2	Welding Boss 3/4" NPT Half Cou SS316 Tank Welding Socket		1 EA
3	Cable Gland M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		1 EA
4	Viton / EPDM Thread Seal Ring Process Sealing Gasket		1 EA
5	Grounding Strap 4mm² with Lugs Copper Flexible Earth Braid		1 EA

ENGINEERING INSTALLATION NOTES

- Install fork with tines aligned vertically to prevent slurry buildup on top of vibrating blades.
- Set fail-safe mode switch: MAX (Overfill Protection) or MIN (Dry Run Pump Protection).
- Keep fork at least 50mm away from tank walls, internal baffles, and agitator blades.

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LIQUID LEVEL SWITCH (VIBRATING FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

DWG NO: KAL-JC-TYP-012
REV: 3.6 DATE: 2026-09-08

PROJECT NO:
R5THS00138-JC

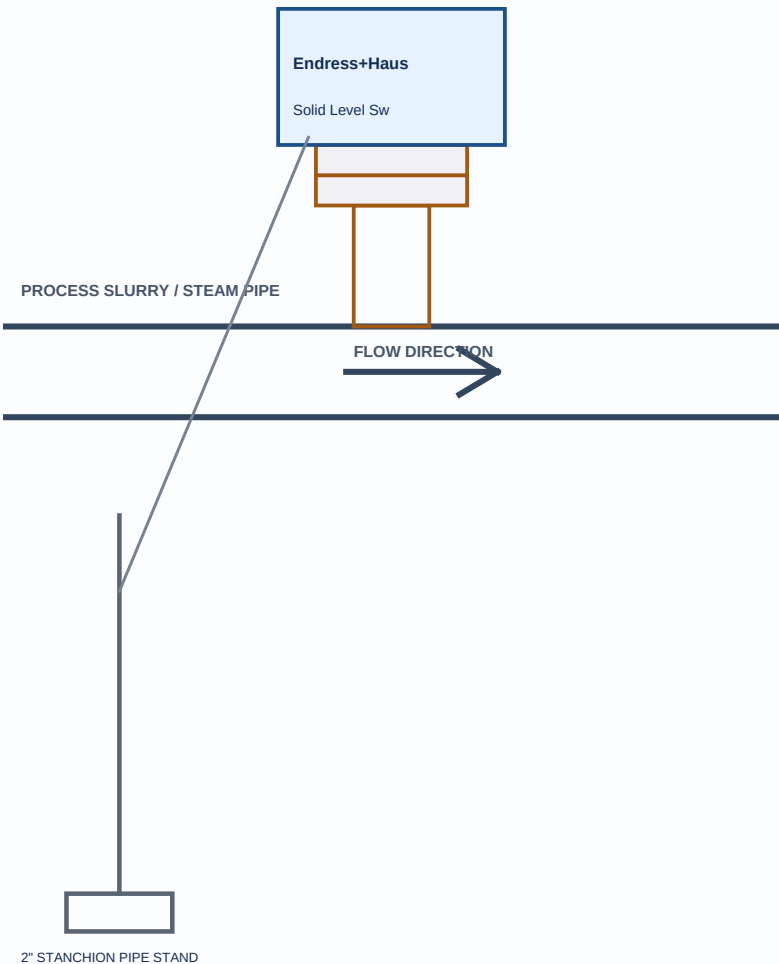
SHEET NO:
14 / 25

TEMPLATE TYP-13: SOLID LEVEL SWITCH

TYPICAL TAGS: LSH-60101 Series

SERVICE: Starch Powder Silo High Level Overfill Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

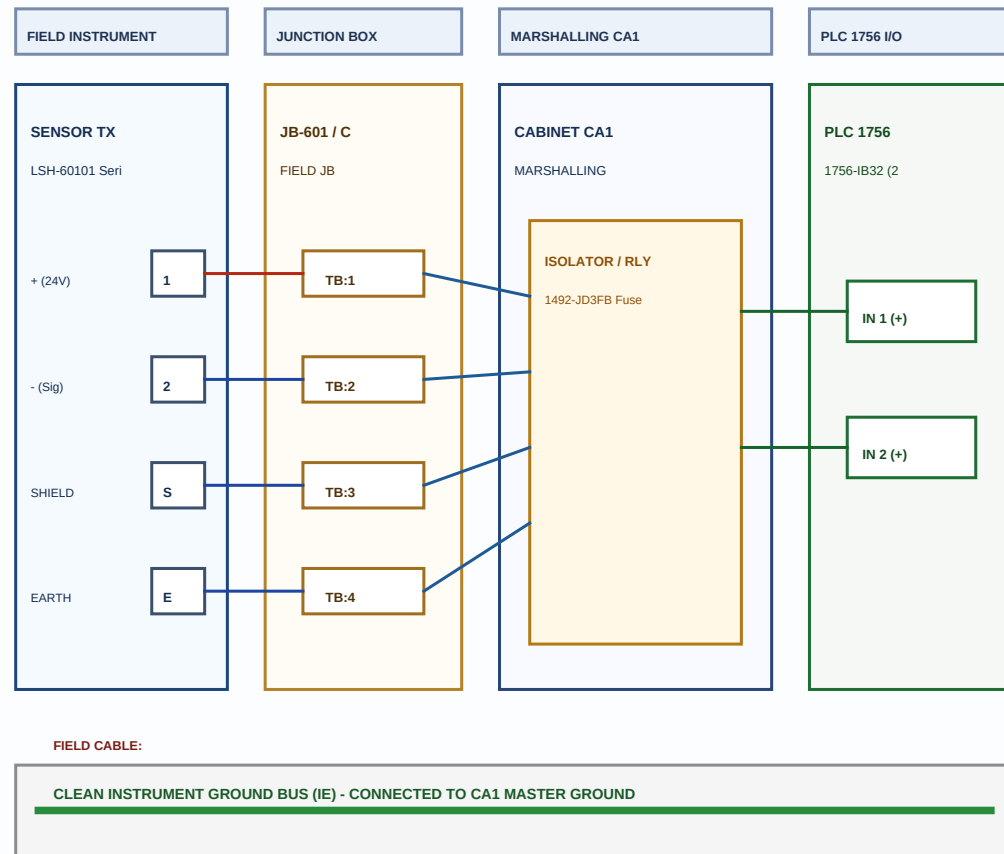
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1-1/2" NPT Male / DN50 Cl.15..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66 Cast Aluminum..
Hazardous Class:	ATEX II 1/2D Ex ta/tb IIIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: Silo Full Alarm Interlock
Power Supply:	24V DC (19-55V DC, 1.5W)
Field Cable Type:	4Cx1.5mm² Shielded Overall (Belden 9534)
Destination JB:	JB-601 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Terminal + Relays
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Soliphant FTM51 (Vibrating Fork for Fine Bulk Solids)

SERVICE: Starch Powder Silo High Level Overfill Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Soliphant FTM51 Powder Level S E+H Fork Extension SS316L (300mm)		1 EA
2	Welding Flange DN50 PN16 / Cl. Carbon Steel / SS316 Flange		1 EA
3	Weather Protective Sun / Rain SS304 Protective Roof		1 EA
4	Dust-Ex Cable Gland M20x1.5 Nickel-plated Brass ATEX Dust GI		1 EA
5	Grounding Wire 6mm² Green/Yell Copper Earthing Strap		1 EA

ENGINEERING INSTALLATION NOTES

- Install angled downward 20° to 30° on silo side wall to allow powder to slide off times easily.
- Do not mount directly in the filling stream or impact zone of falling starch powder.
- Ensure housing earth terminal is connected to silo metal structure to dissipate electrostatic charge.

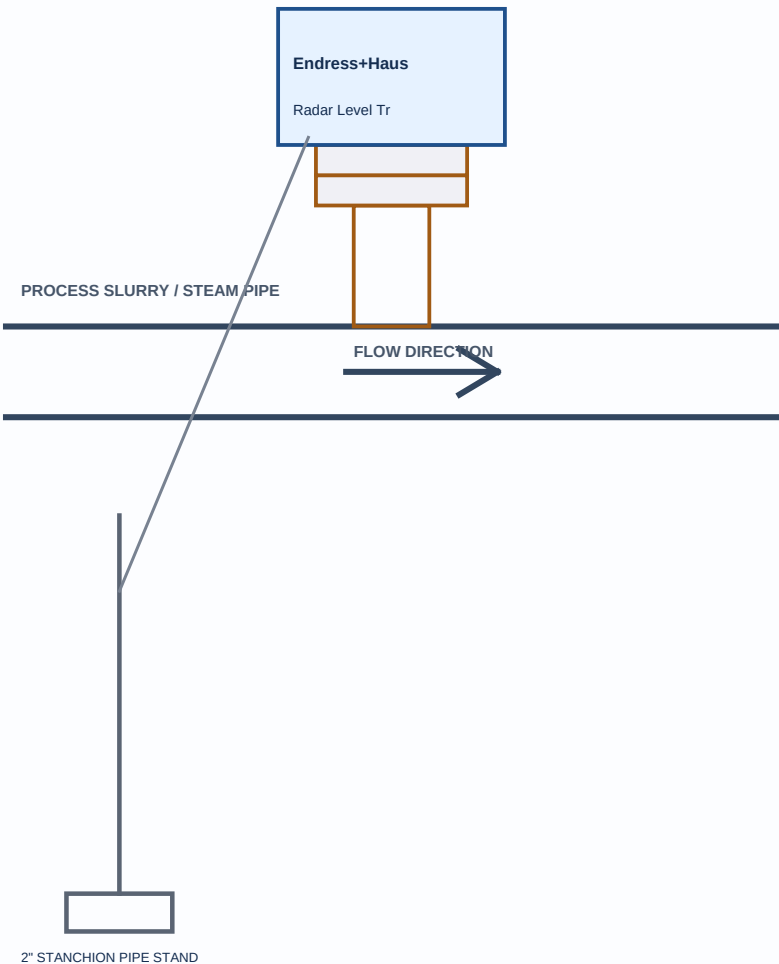
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONSOLID LEVEL SWITCH (POWDER FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-013
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
15 / 25

TEMPLATE TYP-14: RADAR LEVEL TRANSMITTER

TYPICAL TAGS: LT-40101 / LT-40201

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level | PLC I/O: 1756-IF16 (16-Point Differential AI)

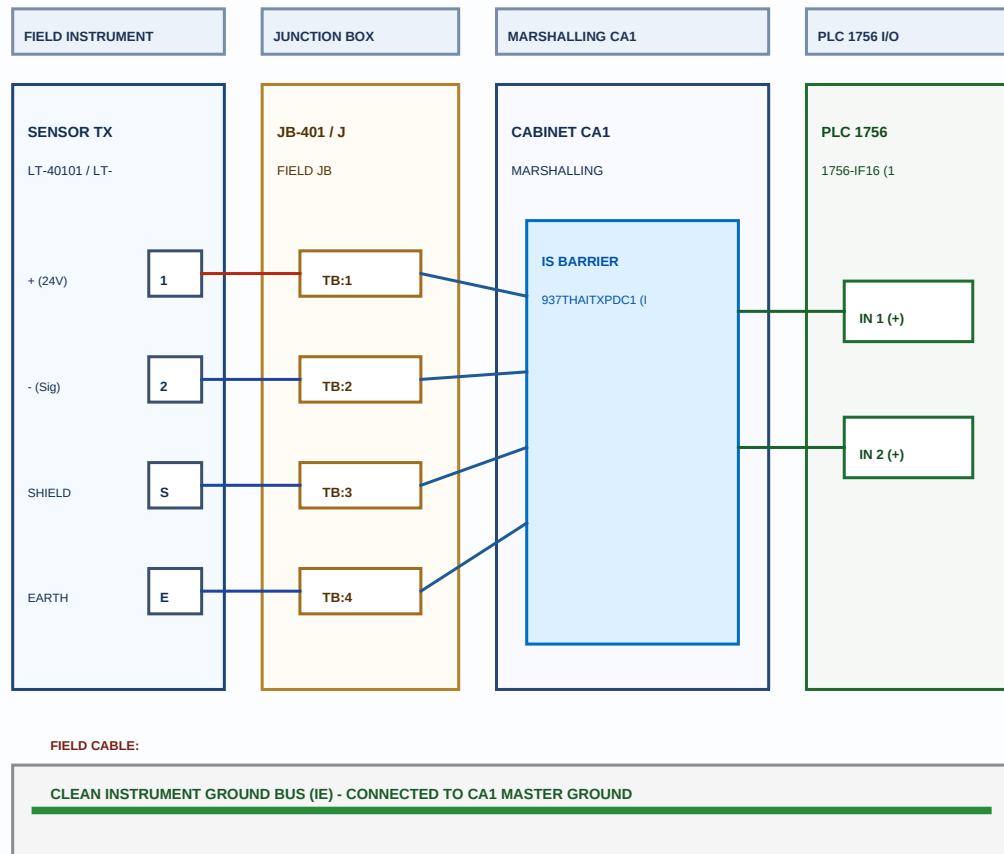
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80 (3") ASME Cl.150 RF Fla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Level (0-100% / 0-6.0m)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Pair Shielded (Belde..
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Micropilot FMR67B (80 GHz High-Frequency Radar)

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Micropilot FMR67B Radar Transm E+H 80GHz w/ Drip-off PTFE Anten		1 EA
2	Tank Nozzle Flange DN80 Cl.150 SS316 Tank Flange Neck		1 EA
3	PTFE Envelope Gasket DN80 Cl.1 Chemical Resistant Gasket		1 EA
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Gland		1 EA
5	Air Purge Connection Kit 1/4" Purge Connection for Condensate		1 SET

ENGINEERING INSTALLATION NOTES

- Mount nozzle height must not exceed antenna length so horn protrudes clear into vessel interior.
- Distance to vessel wall should be min. 1/6 of vessel diameter to prevent false sidewall echoes.
- Activate Heartbeat Technology and mapping curves to mask agitator blade echoes during commissioning.

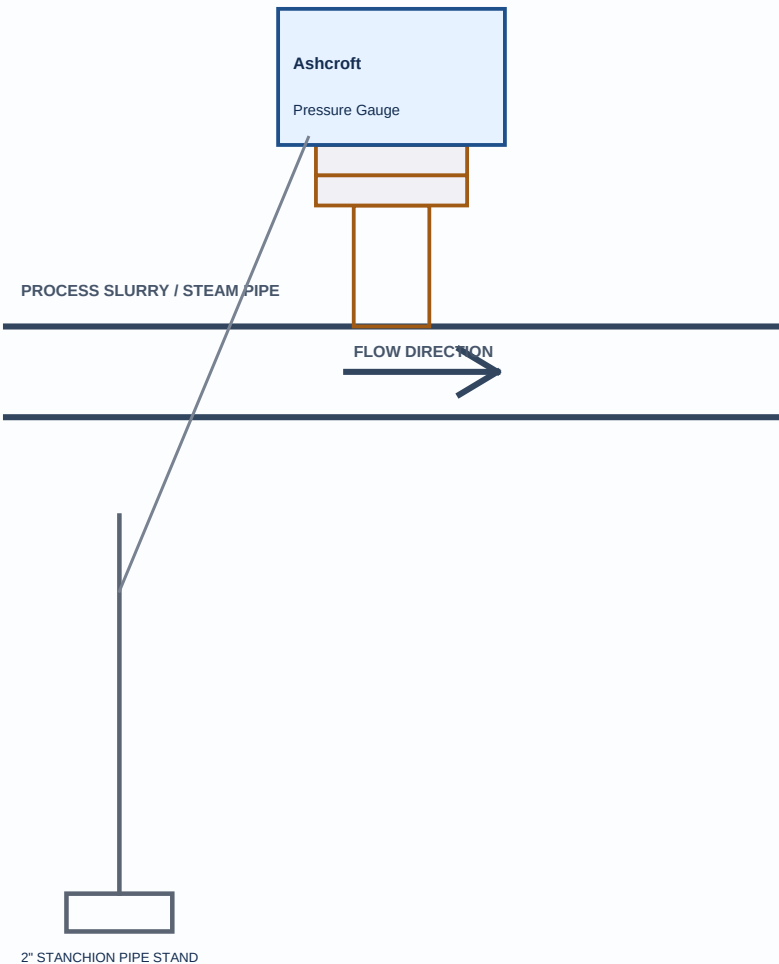
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONRADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-014
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
16 / 25

TEMPLATE TYP-15: PRESSURE GAUGE

TYPICAL TAGS: PI-40101 / PI-60201

SERVICE: Steam Line & Cooker Chamber Direct Pressure Readout | PLC I/O: None (Local Operator Gauge)

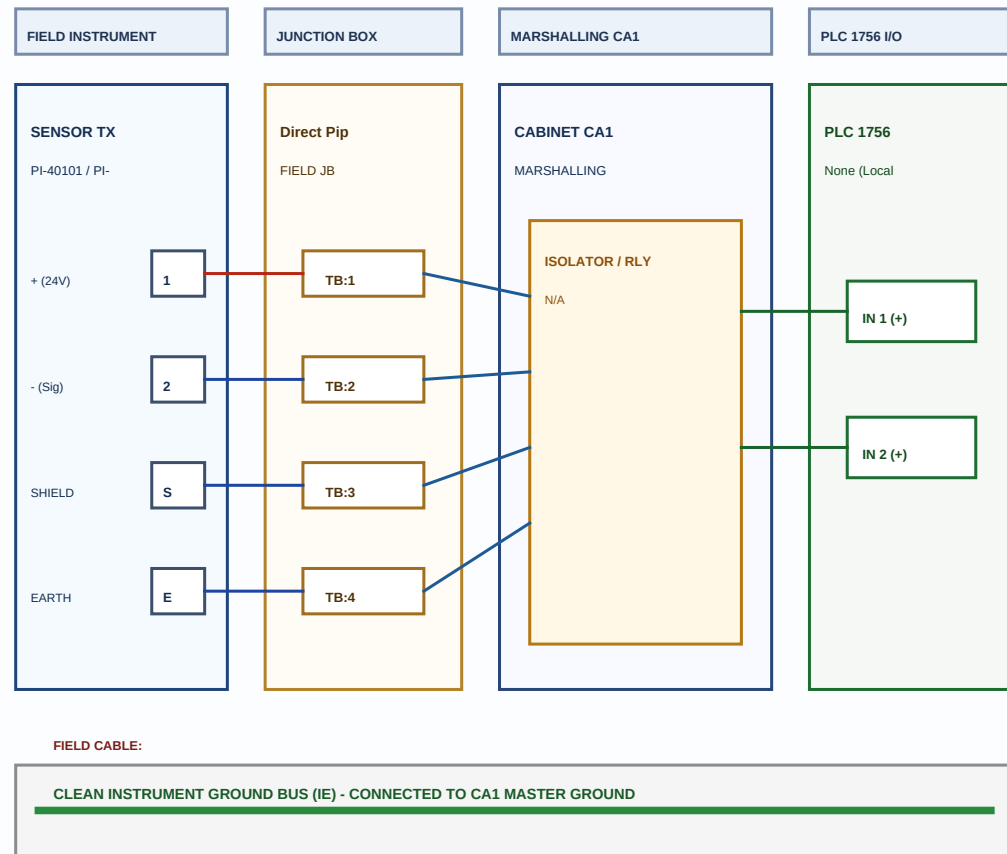
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male w/ Pigtail Siphon
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 Stainless Steel Case, G..
Hazardous Class:	Non-Electrical Apparatus (St..)
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	Visual Dial Scale 0-16 bar (0-230 psi)
Power Supply:	None (Bourdon Tube Mechanical)
Field Cable Type:	N/A (Direct Local Readout)
Destination JB:	Direct Piping Mount
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	N/A
PLC Module:	None (Local Operator Gauge)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ASHCROFT

MODEL: Model T5500 (100mm Dial All-Stainless Steel)

SERVICE: Steam Line & Cooker Chamber Direct Pressure Readout

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Pressure Gauge T5500 (0-16 bar Ashcroft 100mm Dial SS316 Bourdon		1 EA
2	Pigtail Siphon Schedule 80 SS3 1/2" NPT Steam Condensate Loop		1 EA
3	Needle Isolation Valve 1/2" NP SS316 Cl.6000 Needle Valve		1 EA
4	Gauge Snubber / Pulsation Damp Porous SS316 Disc 1/2" NPT		1 EA
5	Thread Sealant Tape (PTFE/Grap High Temperature Steam Rated		1 ROLL

ENGINEERING INSTALLATION NOTES

- Pigtail siphon must be pre-filled with clean water prior to steam startup to protect Bourdon tube.
- Orient dial facing primary walkway / operating platform for ease of operator reading.
- Close needle valve slightly during excessive pulsating flow to protect internal linkage.

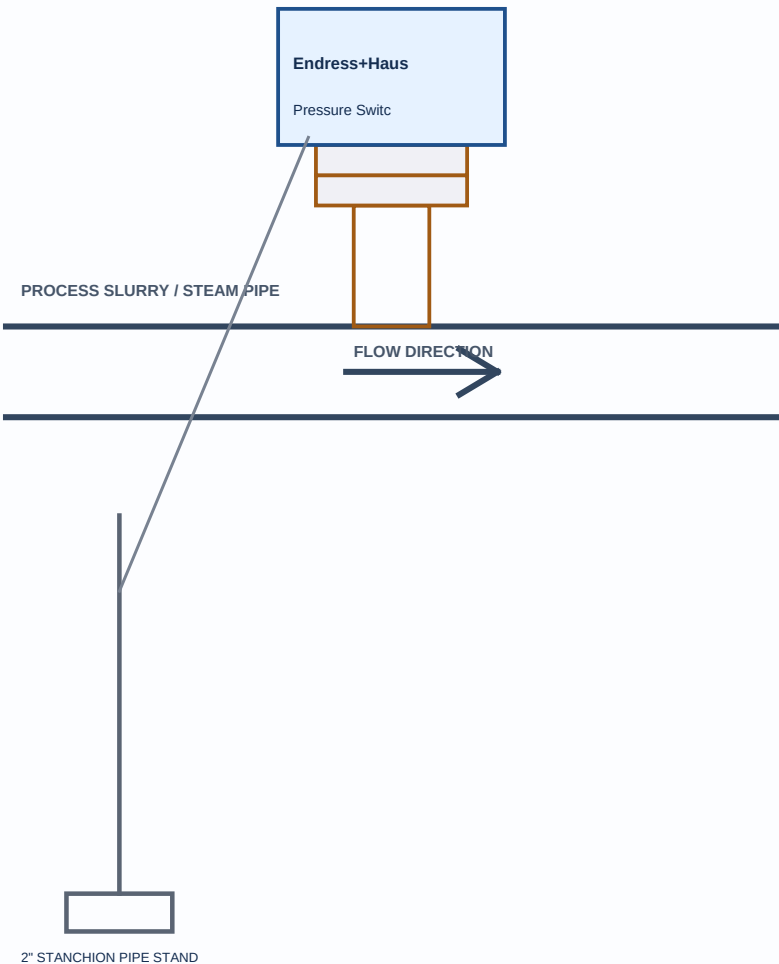
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTPRESSURE GAUGE & SIPHON HOOK-UP & MECHANICAL DETAIL
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-015
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
17 / 25

TEMPLATE TYP-16: PRESSURE SWITCH

TYPICAL TAGS: PSH-60201 / PSL-60201

SERVICE: High / Low Pressure Safety Trip Interlock | PLC I/O: 1756-IB32 (24V DC Digital Input)

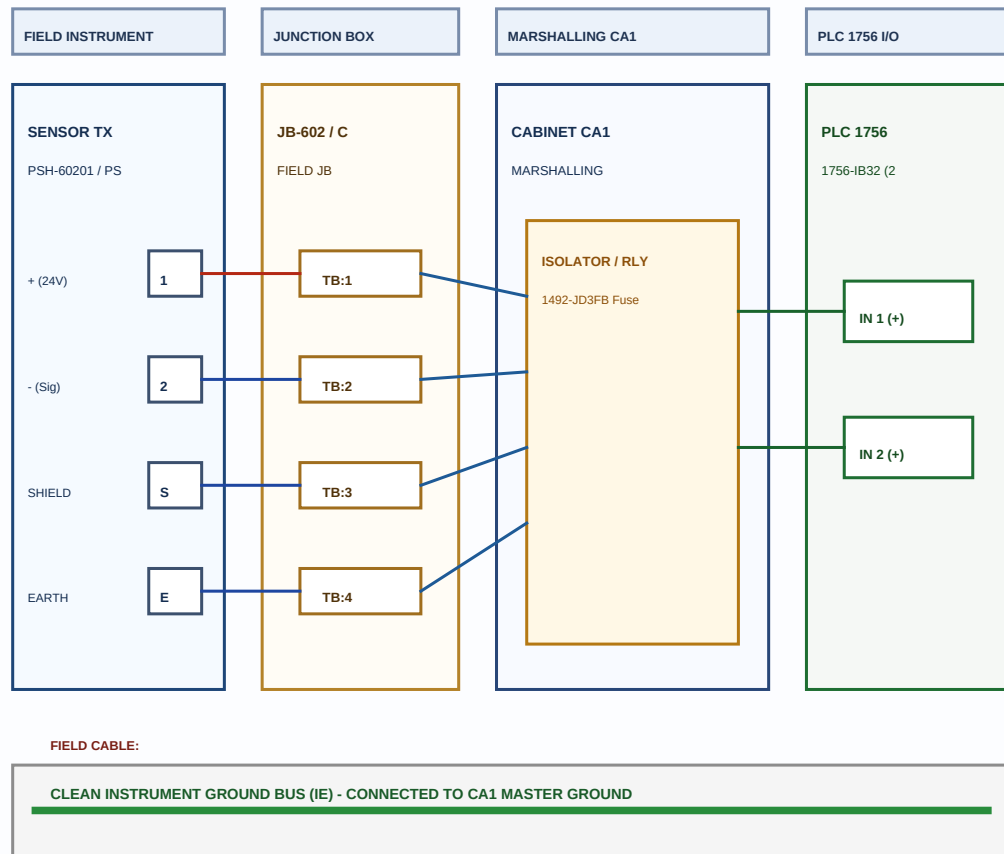
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial Zone (Opt..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: Trip Contact (Open on Alarm)
Power Supply:	24V DC (10-30V DC, 20mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrumentation Cable
Destination JB:	JB-602 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Ceraphant PTC31B (Ceramic Sensor Diaphragm)

SERVICE: High / Low Pressure Safety Trip Interlock

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Ceraphant PTC31B Pressure Swit E+H Ceramic Cell, LED Display		1 EA
2	2-Valve Isolation Manifold 1/2 SS316 Block & Bleed Valve		1 EA
3	M12 4-Pin Field Wireable Conne IP67 Female M12 Connector		1 EA
4	Cable Gland M16x1.5 Brass Nickel-plated Cable Gland		1 EA
5	Impulse Adapter 1/2" NPT x 1/2 Swagelok SS316 Tube Fitting		1 EA

ENGINEERING INSTALLATION NOTES

- Program switch point and reset hysteresis via front push-buttons and LED display.
- Configure output as Normally Closed (NC) fail-safe: circuit de-energizes on overpressure or power loss.
- Provide block-and-bleed valve for routine verification with field calibrator.

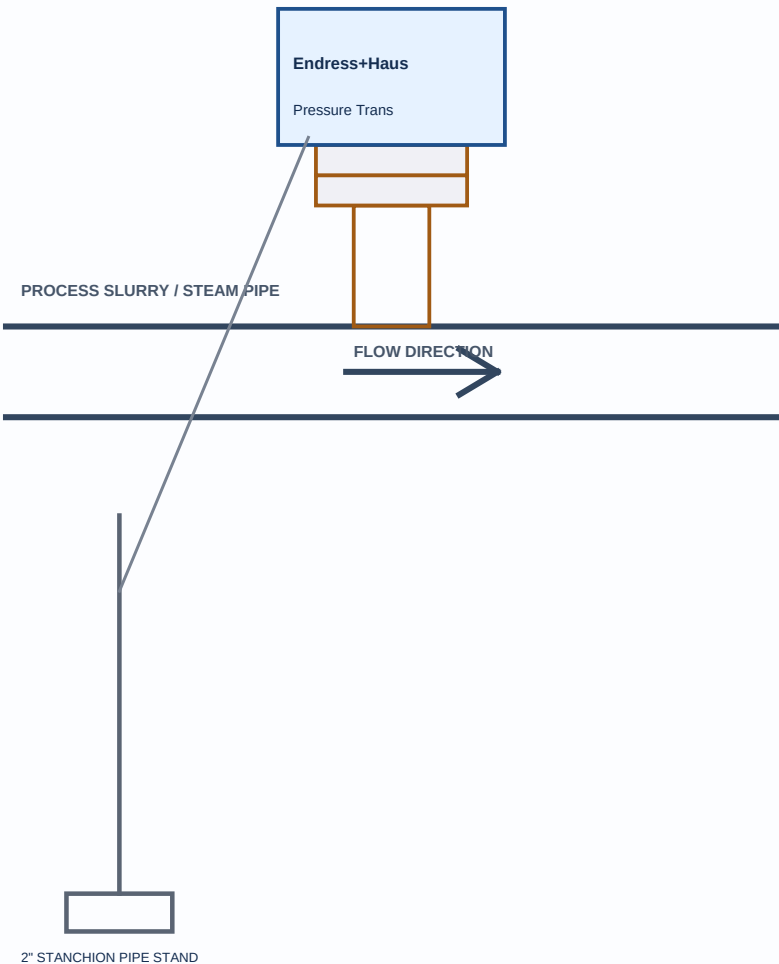
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTRONIC PRESSURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-016
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
18 / 25

TEMPLATE TYP-17: PRESSURE TRANSMITTER

TYPICAL TAGS: PT-40101 / PT-60202

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank | PLC I/O: 1756-IF16 (16-Point Differential AI)

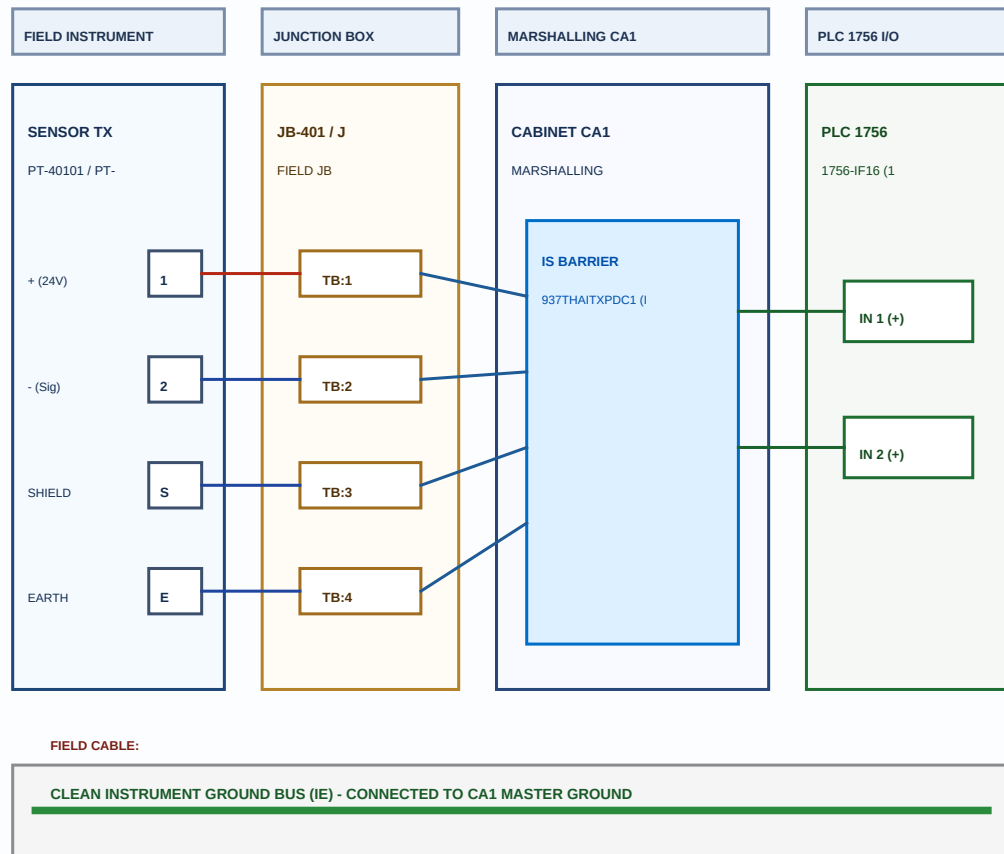
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread / 2" Tr..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



FIELD CABLE:

CLEAN INSTRUMENT GROUND BUS (IE) - CONNECTED TO CA1 MASTER GROUND

ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-10 bar / 0-16 bar gauge)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 8761)
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Cerabar PMP51B / PMC51B (Metallic / Ceramic)

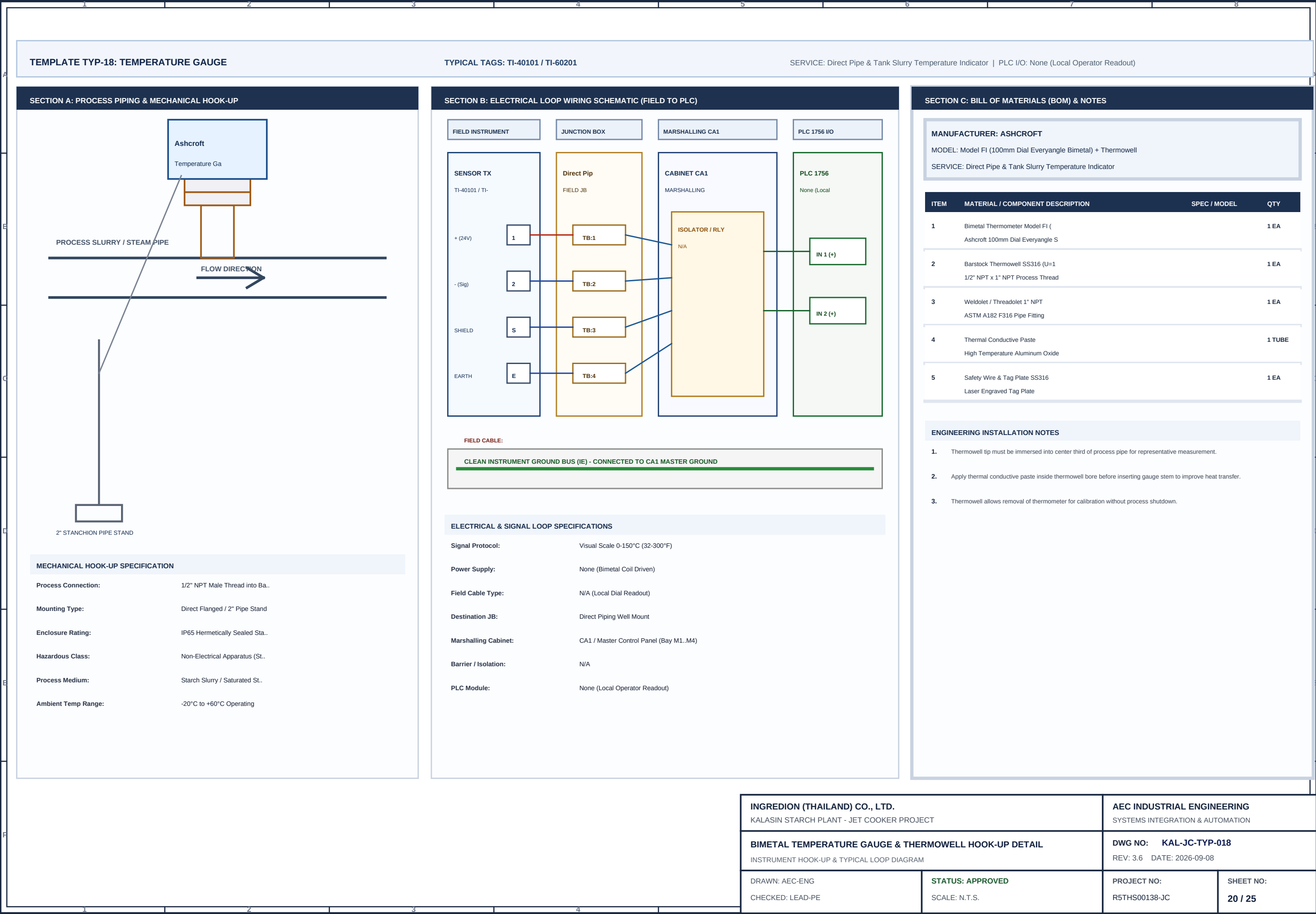
SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank

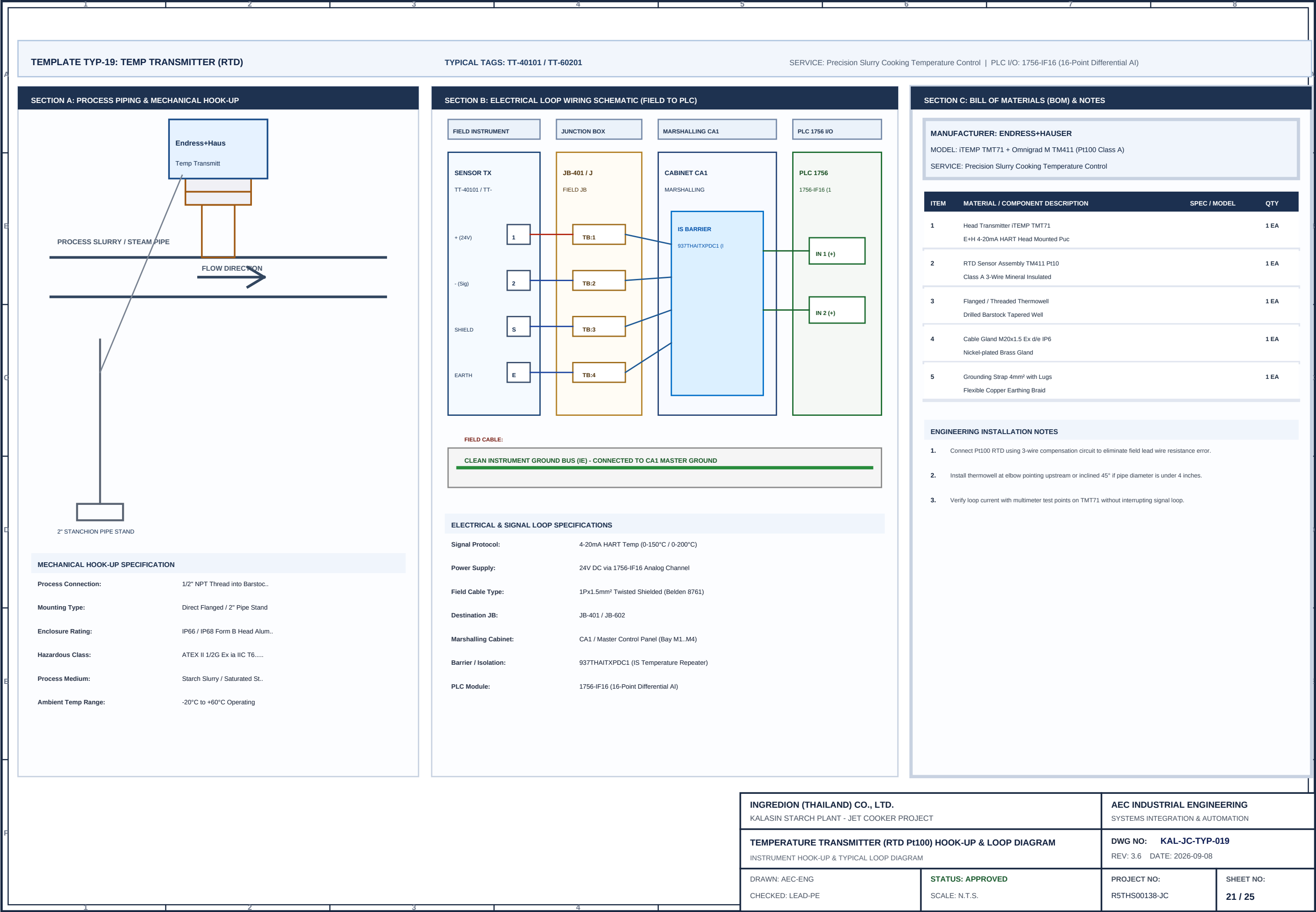
ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Cerabar PMP51B Pressure Transm E+H Piezoresistive SS316L Cell		1 EA
2	2-Valve Block & Bleed Manifold SS316 1/2" NPT Male x Female		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Universal Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Armored Gland		1 EA
5	Siphon / Cooling Neck (Steam D SS316 Pigtail Loop 1/2" NPT		1 EA

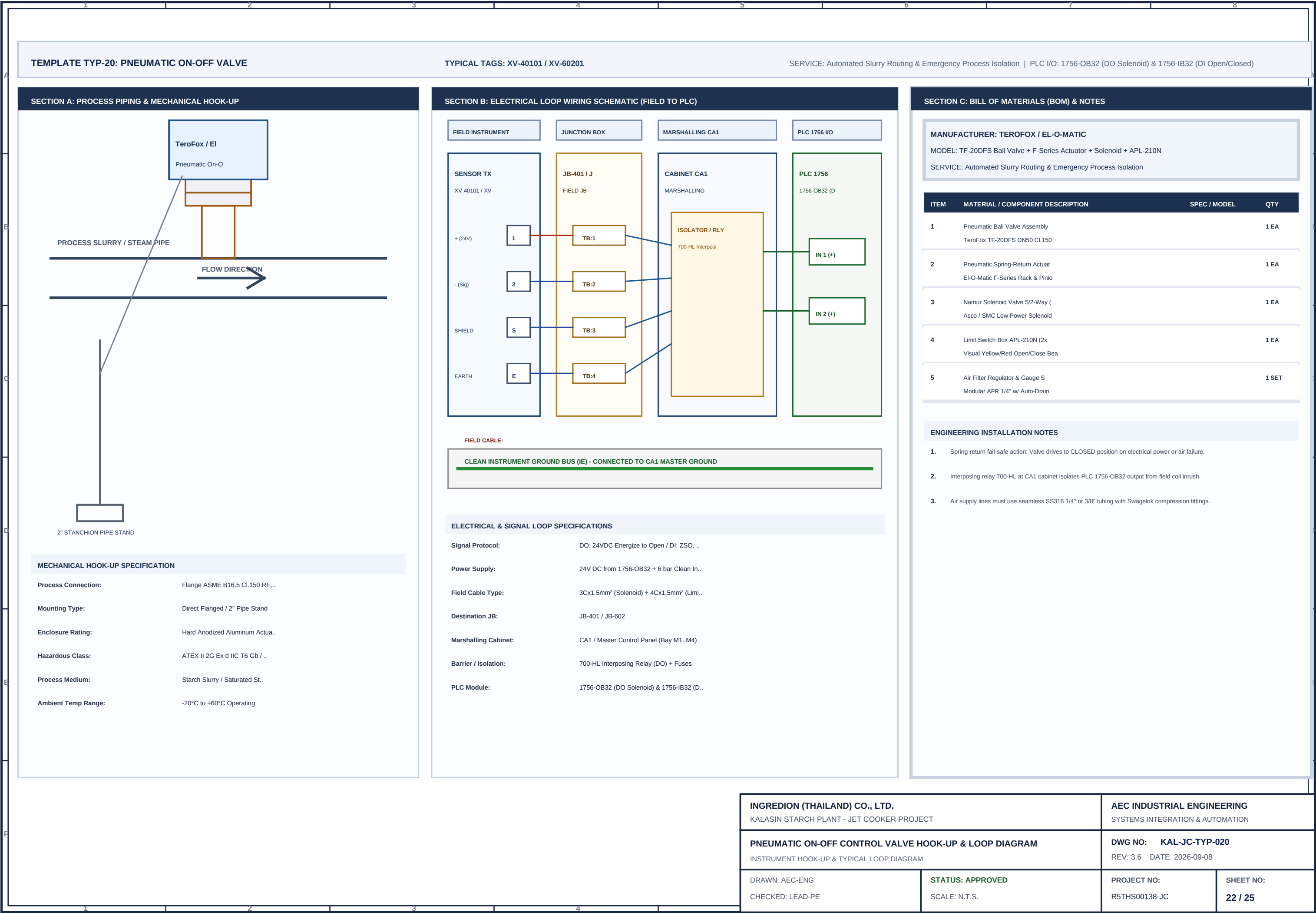
ENGINEERING INSTALLATION NOTES

- Use flush diaphragm seal (PMC51B) for thick starch slurry to eliminate clogging in impulse ports.
- Ground transmitter body to local pipe ground; shield wire grounded only at CA1 cabinet earth bar.
- HART configuration: Set damping to 1.0s to filter process pump pulsations.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONPRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-017
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
19 / 25





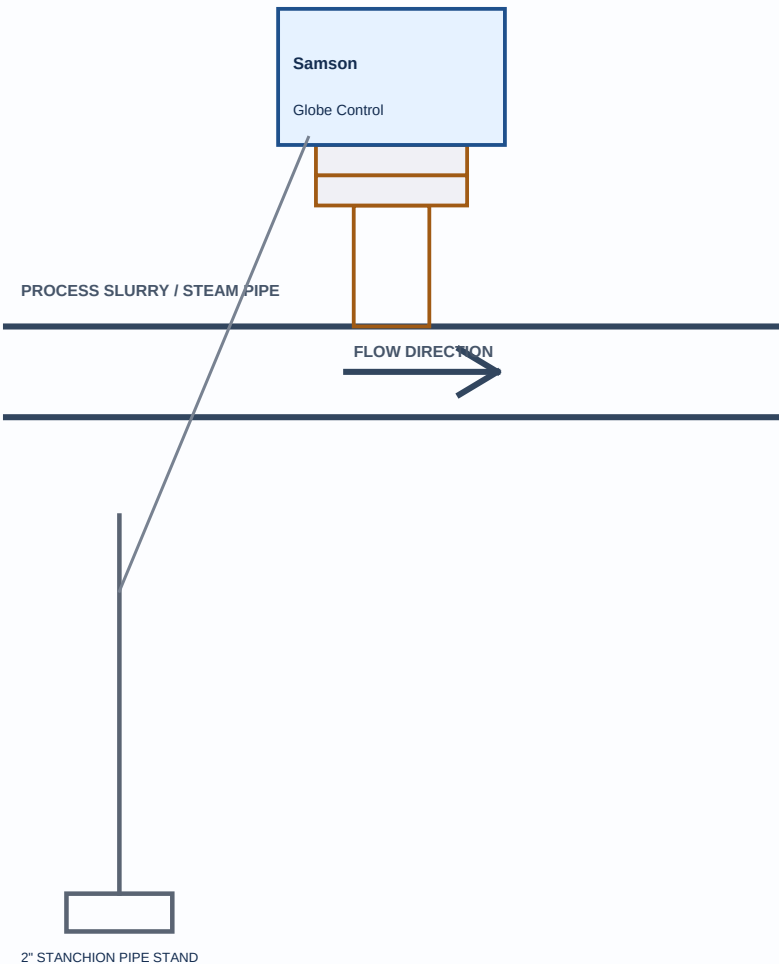


TEMPLATE TYP-21: GLOBE CONTROL VALVE

TYPICAL TAGS: TCV-40222 / TV-60202

SERVICE: High-Precision Steam & Slurry Flow Modulation | PLC I/O: 1756-OF8 (AO Setpoint) & 1756-IF16 (AI Feedback)

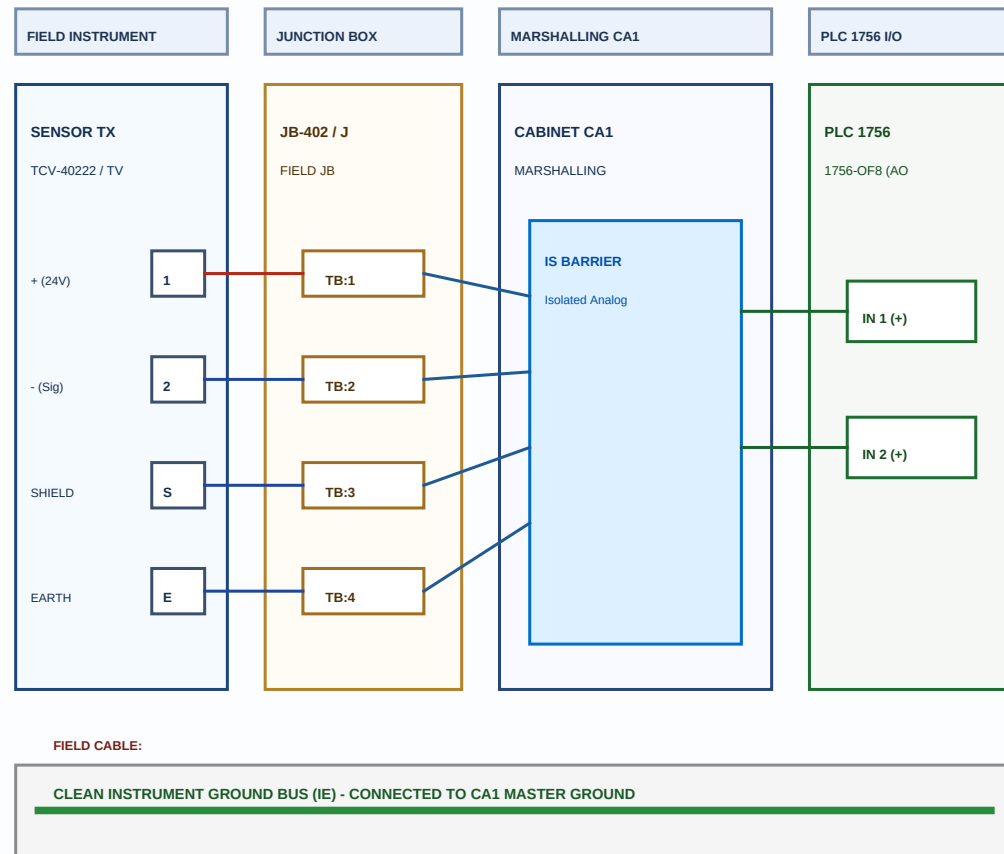
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	Flange ASME B16.5 Cl.150/300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP66 Die-cast Aluminum Posit..
Hazardous Class:	ATEX II 2G Ex ia IIC T6 / Ex..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AO: 4-20mA Valve Cmd (0-100%) / AI: Po..
Power Supply:	2-Wire 4-20mA Loop + 6.0 bar Instrumen..
Field Cable Type:	2Px1.5mm ² Twisted Shielded (Belden 8777)
Destination JB:	JB-402 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Isolated Analog Output / IS Repeater
PLC Module:	1756-OF8 (AO Setpoint) & 1756-IF16 (AI..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: SAMSON

MODEL: Type 3241 Globe Valve + Trovis 3730-1 Smart Positioner

SERVICE: High-Precision Steam & Slurry Flow Modulation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Globe Control Valve Type 3241 Samson Cast Steel / SS316 Body		1 EA
2	Pneumatic Diaphragm Actuator T Samson Multi-Spring Actuator		1 EA
3	Smart Positioner Trovis 3730-1 HART Communication, Auto-Tuning		1 EA
4	SMC Air Filter Regulator AW20 Clean Air 1/4" w/ Pressure Gauge		1 SET
5	Cable Glands M20x1.5 Ex d/e IP Nickel-plated Brass Glands		2 EA

ENGINEERING INSTALLATION NOTES

- Perform Trovis 3730-1 automatic initialization routine after mechanical mounting is completed.
- Connect 4-20mA setpoint from PLC 1756-OF8 to Positioner Terminals 11(+) and 12(-).
- Ensure air supply pressure to positioner is regulated strictly between 4.0 and 6.0 bar.

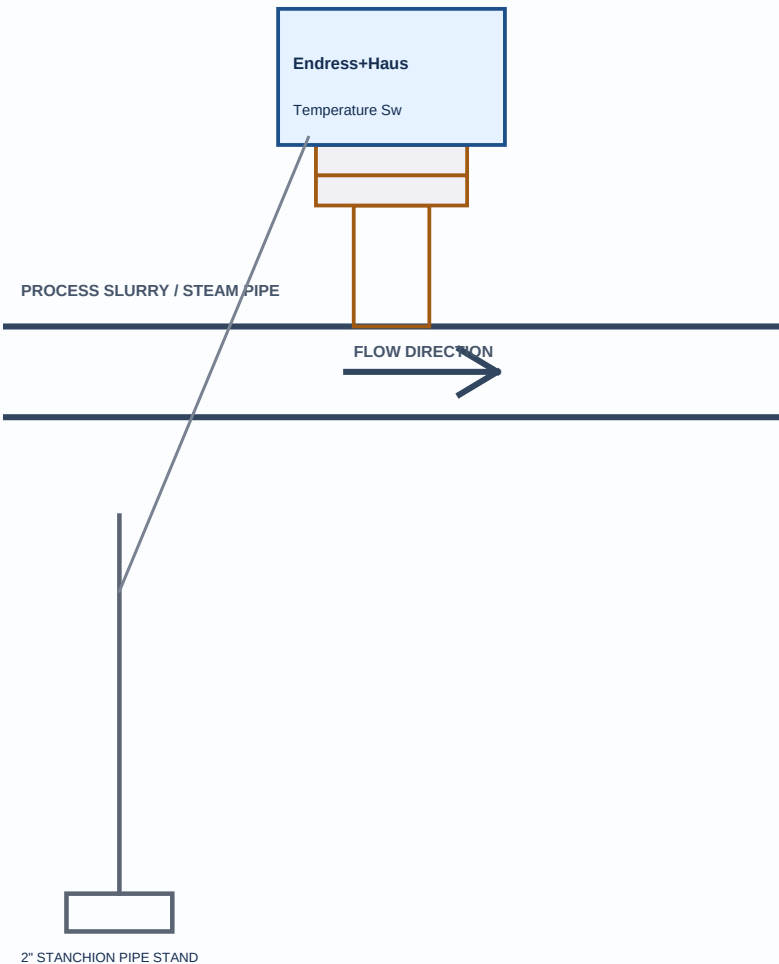
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONMODULATING GLOBE CONTROL VALVE HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-021
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
23 / 25

TEMPLATE TYP-22: TEMPERATURE SWITCH

TYPICAL TAGS: TSH-40101 / TSH-60201

SERVICE: High Temperature Safety Trip & Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

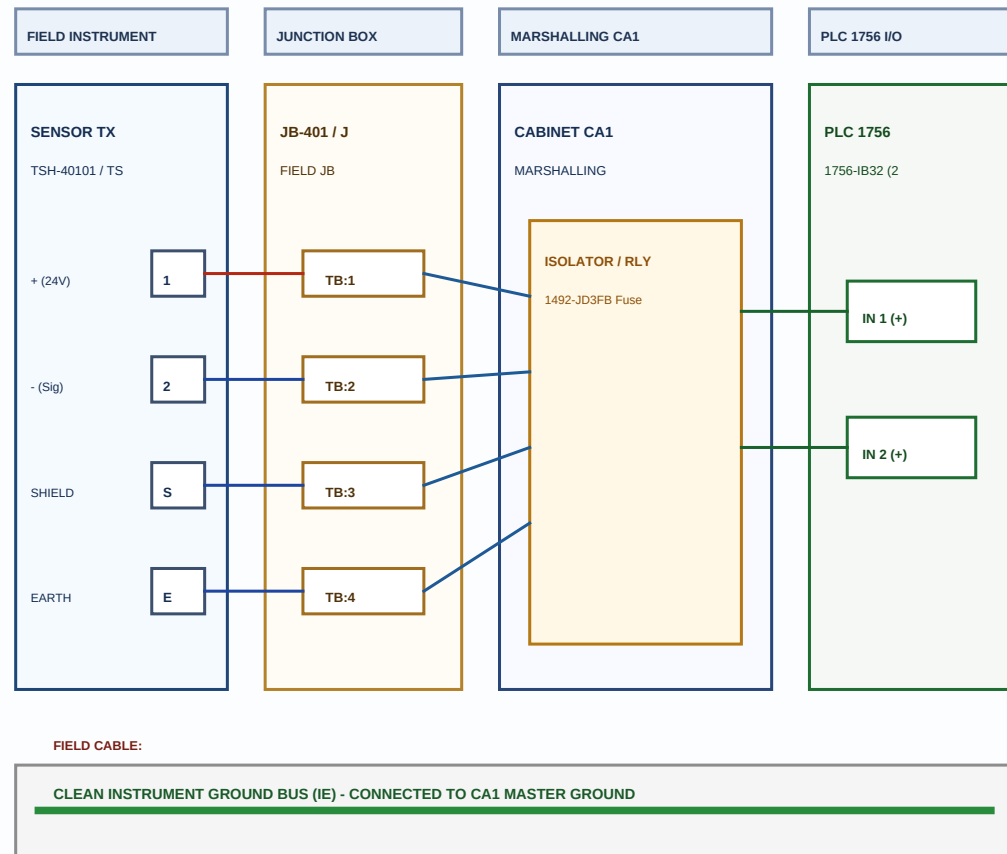
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Hyg..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial / Hygieni..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Temperature Alarm Trip
Power Supply:	24V DC (10-30V DC, 35mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Thermophant TTR35 (Compact Pt100 Switch)

SERVICE: High Temperature Safety Trip & Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Thermophant TTR35 Temp Switch E+H Pt100 Class A, Digital Displ		1 EA
2	Barstock Thermowell SS316L (U= 1/2" NPT x 1/2" NPT Process Well		1 EA
3	M12 4-Pin Female Cable Assembl M12 Right Angle Molded Connector		1 EA
4	Thredolet / Boss 1/2" NPT SS3 Pipe Welding Fitting		1 EA
5	Grounding Clamp & Lug Copper Earth Wire		1 SET

ENGINEERING INSTALLATION NOTES

- Configure switch point via optical touch keys on front cover without opening enclosure.
- Normally closed (NC) contact output configuration guarantees fail-safe alarm on wire break.
- Thermowell allows switch replacement without draining the process line.

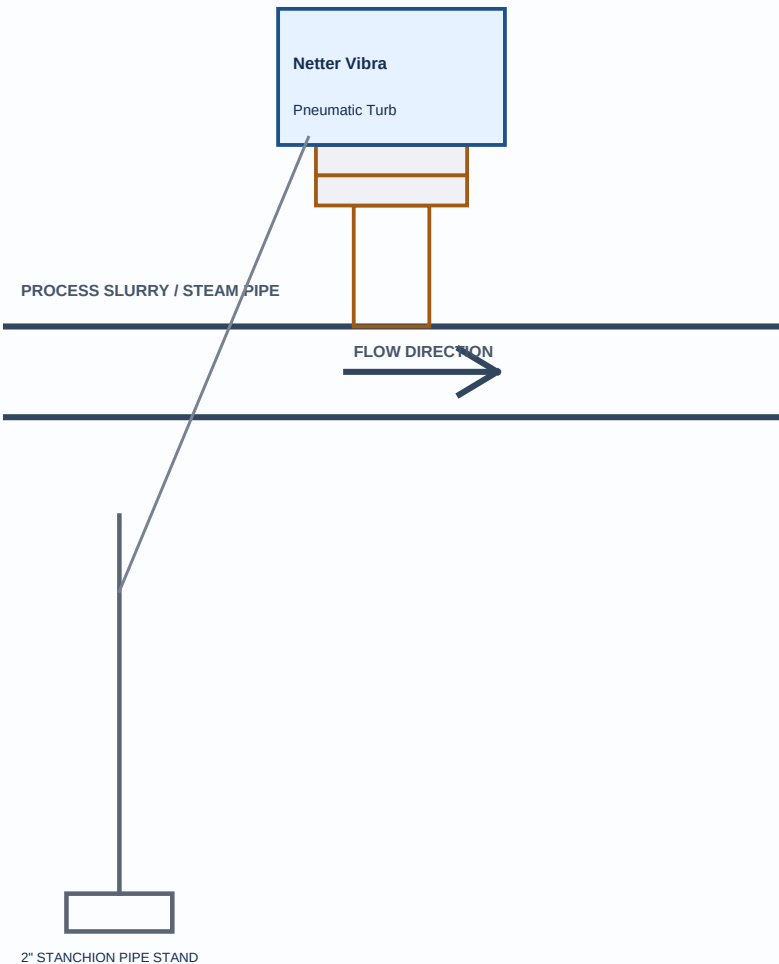
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTRONIC TEMPERATURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-022
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
24 / 25

TEMPLATE TYP-23: PNEUMATIC TURBINE VIBRATOR

TYPICAL TAGS: XV-60901 Series

SERVICE: Powder Silo & Hopper Wall Discharge Agitation | PLC I/O: 1756-OB32 (24V DC Digital Output)

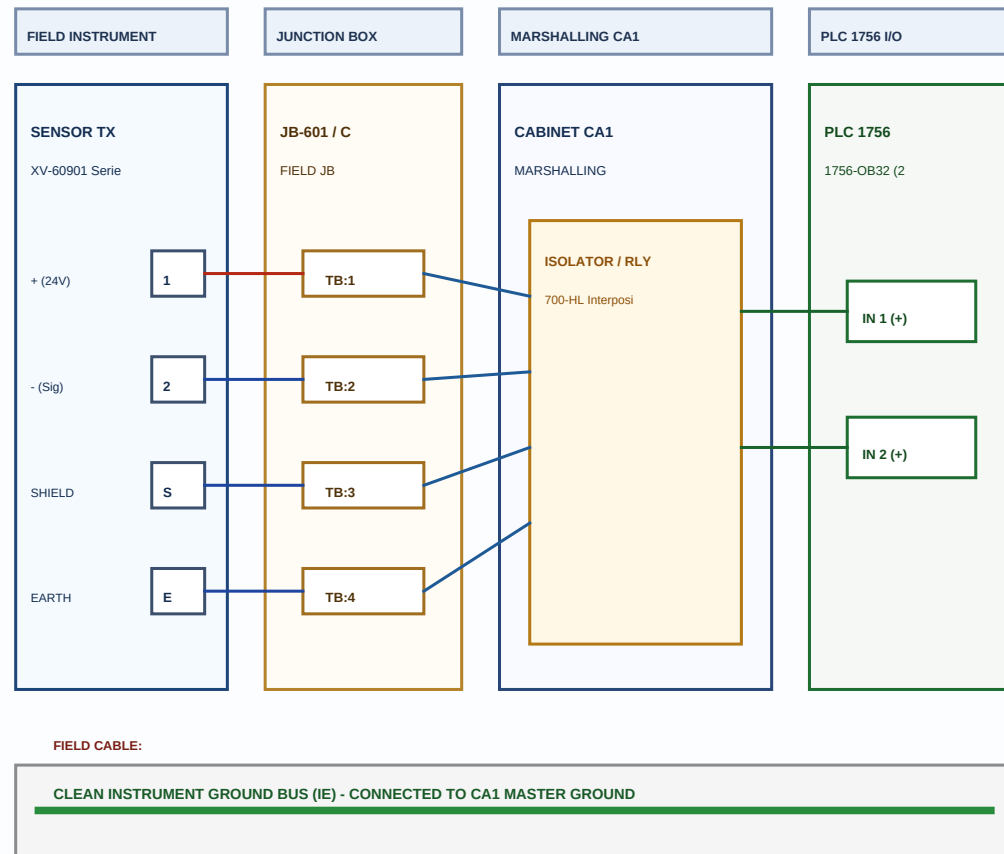
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	G 1/8" BSP Air Port / Rigid ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Extruded Aluminum Hard Anodi..
Hazardous Class:	ATEX II 2GD c T6 85°C (Mecha..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DO: Pulsed Vibration Burst (Timer Cont..
Power Supply:	24V DC from 1756-OB32 + 2 to 6 bar Lub..
Field Cable Type:	3Cx1.5mm² Shielded Cable (Belden 9533)
Destination JB:	JB-601 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	700-HL Interposing Relay with Diode
PLC Module:	1756-OB32 (24V DC Digital Output)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: NETTER VIBRATION

MODEL: NCT 5 Pneumatic Turbine Vibrator

SERVICE: Powder Silo & Hopper Wall Discharge Agitation

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Pneumatic Turbine Vibrator NCT Netter Vibration High-Speed Unit		1 EA
2	Solenoid Valve 3/2-Way 24VDC (High Flow Fast-Cycling Solenoid		1 EA
3	Silencer / Exhaust Air Filter Sintered Bronze Sinter Filter		1 EA
4	SMC Air Filter Regulator Kit Modular AFR w/ Pressure Gauge		1 SET
5	Weld-on Hopper Mounting Channe Reinforced Channel Profile Steel		1 EA

ENGINEERING INSTALLATION NOTES

- Mount vibrator on stiffened channel beam welded to hopper cone to transmit vibration into bulk powder.
- Operate in intermittent pulse mode (e.g. 5 seconds ON, 30 seconds OFF) via PLC timer logic.
- Interposing relay 700-HL at CA1 panel protects PLC output module from inductive flyback.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONPNEUMATIC TURBINE VIBRATOR HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-023
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
25 / 25